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AquaYOLO26: A Degradation-Aware YOLO26 Framework for Underwater Marine Debris Detection and Edge Deployment

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Abstract

Against the backdrop of escalating marine pollution, automated visual detection of submerged debris by autonomous underwater vehicles (AUVs) is essential for robotic ocean cleanup. However, robust underwater detection is fundamentally challenged by an asymmetric visual degradation process: wavelength-dependent light attenuation weakens red channel responses more severely than blue–green channels, turbidity-induced scattering disrupts the symmetry of small target features, and cross-water-domain shifts further distort feature distributions. To address these challenges, this paper proposes AquaYOLO26, a degradation-aware and domain-adaptive edge detection framework that explicitly models underwater asymmetry while restoring feature-level balance for robust perception. Specifically, we introduce Underwater-aware Batch Normalization (UWBN) to compensate for channel-asymmetric color attenuation within the feature extractor; Turbidity-conditioned Small Target-Aware Label Assignment (Turb-STAL) to asymmetrically adjust candidate assignment for obscured small targets under scattering conditions; and Domain-Adversarial Progressive Loss (DA-ProgLoss) to align feature representations across heterogeneous water environments. Experiments on the SeaClear, TrashCan 1.0, and Trash-ICRA19 benchmarks show that AquaYOLO26n achieves 73.4 $\pm$ 0.33% mAP@0.5 on the primary SeaClear test split and, in the representative run cross-domain evaluation, improves strict zero-shot transfer performance on Trash-ICRA19 by 8.9 percentage points over the YOLO26n baseline. For embedded deployment, the model retains a compact profile of 2.46 M parameters and 6.2 GFLOPs, achieving 45.1 $\pm$ 0.5 FPS on the NVIDIA Jetson Orin NX. Compared with Improved YOLOv11, AquaYOLO26n achieves a 1.1 pp higher mean mAP@0.5 while using 79.0% fewer parameters and 77.0% fewer GFLOPs and delivering higher Jetson throughput (45.1 $\pm$ 0.5 vs. 28.4 $\pm$ 0.5 FPS). These results highlight a favorable accuracy–efficiency trade-off for resource-constrained underwater robotic sensing.

Keywords: underwater object detection; marine debris; YOLO26; channel-asymmetric degradation; feature symmetry restoration; degradation-aware detection; domain adaptation; zero-shot transfer; edge deployment

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1. Introduction

Marine debris pollution has become one of the most urgent threats to global ocean health. An estimated 4.8 to 12.7 million metric tons of plastic waste enter the world's oceans

each year [1], and cumulative floating plastic concentrations exceed 250,000 tons across the major oceanic gyres [2]. Beyond surface accumulation, submerged debris—including discarded fishing nets, metallic fragments, glass, and synthetic rope—settles on the seafloor and in the water column, causing long-term damage to benthic ecosystems, marine biodiversity, and coastal fisheries [3,4]. The rapid growth of global plastic production and waste accumulation further intensifies the need for scalable underwater debris monitoring and removal technologies [5]. To mitigate this environmental crisis, engineering interventions for debris localization, collection, and systematic removal are urgently needed. However, whether retrieval is performed by surface vessels, remotely operated vehicles, or robotic manipulators, physical recovery cannot proceed without reliable visual perception. Robust real-time underwater object detection is therefore a prerequisite and a key technological bottleneck for automated marine cleanup.

Researchers and international consortia have developed diverse computer vision methods to address this challenge. For example, the European Commission's Horizon 2020 program funded the SeaClear consortium [6], which pioneered cooperative robotic systems using autonomous underwater vehicles (AUVs) for seafloor litter mapping. Researchers at the University of Minnesota also released the TrashCan [7] and Trash-ICRA19 [8] datasets, substantially advancing deep learning applications in marine robotics. These efforts have addressed specific visual challenges: SEANet [9] introduced multiscale detail amplification and foreground–background contrast attention, and attention-based frameworks such as YOLO-GE [10] optimized feature fusion for aquatic targets. Other representative contributions include AGS-YOLO [11] for lightweight small target detection, MD-YOLO [12] for imbalanced target aspect ratios, IFEM-YOLOv13 [13] for feature symmetry balancing, and UDD-YOLO [14] for integrating cold diffusion models with object detectors. More recently, Zhou and Ma proposed EFE_UODNet, which combines enhanced global context-guided feature extraction, multi-scale feature fusion, and the Dynamic Head (DyHead) to improve underwater organism detection in complex scenes [15].

Representative underwater image enhancement (UIE) methods, including Water-Net, UWCNN, FUnIE-GAN, and diffusion-based restoration approaches, have been developed to improve underwater image visibility and color quality [16–19]. However, improvements in perceptual image quality do not necessarily translate into better downstream vision performance, as reported in underwater feature-matching tasks [20]. Moreover, an independent enhancement model introduces an additional processing stage, which may be undesirable for latency-sensitive AUV/ROV deployment. These considerations motivate a detection-integrated solution in which degradation compensation is embedded directly into the feature extractor and jointly optimized with the detection objective.

Against this background, the recently released YOLO26 model family [21] is especially relevant for edge deployment. YOLO26 addresses a critical post-processing bottleneck on autonomous platforms by adopting a native Non-Maximum Suppression (NMS)-free pipeline, eliminating up to 43% of post-processing latency on embedded CPUs [21,22]. Its Small Target-Aware Label Assignment (STAL) mechanism further improves localization of compact targets, making YOLO26 a strong candidate for resource-constrained marine robotic platforms.

Despite these advances, state-of-the-art detectors still degrade substantially when deployed in real, unpredictable aquatic environments. Water exhibits strong wavelength-dependent absorption; physical oceanographic studies show that red spectrum light is almost completely attenuated within the first 2 to 5 m of water depth [23], producing a dominant blue–green color cast and weakening red channel visual cues. Suspended particulates additionally induce severe forward- and backscattering [24], blurring object boundaries and suppressing high-frequency feature responses. As a result, standard

label assignment strategies, including YOLO26's STAL, may assign background noise as positive anchor candidates in high-turbidity conditions. Finally, large distribution gaps between shallow coastal waters and deep water ROV environments produce severe domain shifts [25], limiting zero-shot field transfer for models trained on a single source dataset.

To address these limitations, we propose AquaYOLO26, a degradation-aware and domain-adaptive framework for underwater marine debris detection. AquaYOLO26 introduces three target-specific designs while preserving the efficient NMS-free inference path of the YOLO26 baseline. First, we replace standard batch normalization with Underwater-aware Batch Normalization (UWBN), which dynamically compensates for wavelength-selective signal loss during feature extraction. Second, we propose Turbidity-conditioned STAL (Turb-STAL), which uses a learned turbidity index to adapt target-matching scores and candidate anchor counts in high-scattering environments. Third, we integrate Domain-Adversarial Progressive Loss (DA-ProgLoss) into the training schedule to align feature representations across heterogeneous water bodies.

To systematically evaluate the proposed framework, this study addresses the following four research questions:

(RQ1) Does AquaYOLO26 improve underwater marine debris detection performance over the YOLO26 baseline and representative underwater detection methods on the primary SeaClear benchmark?

(RQ2) Does the proposed domain-adaptive training framework improve cross-water-domain generalization under unsupervised domain adaptation using unlabeled TrashCan 1.0 images and strict zero-shot transfer to Trash-ICRA19?

(RQ3) What are the individual contributions of UWBN, Turb-STAL, and DA-ProgLoss, and does their integration provide complementary rather than redundant improvements?

(RQ4) Does AquaYOLO26 maintain real-time inference throughput on AUV/ROV-grade embedded hardware, and what inference overhead is introduced relative to the YOLO26 baseline?

Experiments on the SeaClear, TrashCan 1.0, and Trash-ICRA19 benchmarks show that AquaYOLO26 improves detection accuracy, recall, small target detection, and cross-water-domain generalization while preserving a lightweight deployment profile. Under the three-seed evaluation, AquaYOLO26n achieves 73.4 $\pm$ 0.33% mAP@0.5 and 69.1 $\pm$ 0.31% recall with only 2.46 M parameters and 6.2 GFLOPs, while maintaining 45.1 $\pm$ 0.5 FPS on the NVIDIA Jetson Orin NX. The practical contribution of the framework therefore lies in its favorable balance among detection performance, cross-domain robustness, and real-time embedded efficiency, rather than in accuracy improvement alone. The individual novelty of the three core modules and the rationale for their complementary integration are summarized as follows:

1. Underwater-aware Batch Normalization (UWBN): Unlike two-stage "enhancement-then-detection" pipelines and ordinary batch normalization, UWBN integrates the Beer–Lambert underwater light attenuation physical prior into backbone normalization layers. It dynamically estimates an imaging-depth proxy from feature statistics and compensates for channel-asymmetric red signal loss end-to-end. Consistent with our inference pipeline, this design removes the need for standalone preprocessing modules and introduces only marginal inference latency. It substantially mitigates the imbalance in multi-channel feature responses throughout backbone feature extraction.

2. Turbidity-conditioned Small Target-Aware Label Assignment (Turb-STAL): The original YOLO26 STAL uses a fixed base candidate count and does not explicitly account for turbidity-dependent suppression of small target responses. We design a weakly supervised, lightweight turbidity estimator that dynamically adjusts small target matching weights and expands candidate anchor pools based on per-image turbidity estimates during training.

This module only participates in training and introduces no inference time overhead. It improves the balance of positive sample assignment for small targets whose responses are suppressed by turbidity.

3. Domain-Adversarial Progressive Loss (DA-ProgLoss): Although YOLO26's native ProgLoss progressively reweights the detection loss components, it does not explicitly address cross-domain distribution shift. DA-ProgLoss introduces a gradient reversal-based domain classification objective whose alignment strength progressively increases during training. Keeping the alignment pressure low during the early training stage allows the detection objective to establish stable representations before stronger domain alignment is applied. DA-ProgLoss therefore complements UWBN and Turb-STAL by addressing cross-domain representation alignment rather than feature compensation or positive sample assignment.

These three modules are designed to address distinct but complementary stages of the detection pipeline: UWBN performs degradation-aware feature compensation during backbone feature extraction, Turb-STAL adjusts training time positive sample assignment under varying turbidity conditions, and DA-ProgLoss aligns feature representations across water domains. Their integration therefore forms a complementary multi-level framework spanning feature extraction, label assignment, and cross-domain representation learning. The ablation study in Section 5.3 shows that each module improves performance over the YOLO26n baseline and that the complete framework achieves the best overall performance among the evaluated variants.

2. Related Work

This section reviews the literature most relevant to AquaYOLO26 across five connected themes: underwater optical degradation and image enhancement; the evolution of YOLO-based object detection; recent advances in underwater object detection; underwater marine debris datasets and their domain-diversity characteristics; and domain adaptation and generalization for object detection.

2.1. Underwater Optical Degradation and Image Enhancement

The physical basis of underwater image degradation is well established. Water selectively absorbs light at different wavelengths according to the Beer–Lambert exponential attenuation law, causing red channel signal loss to dominate beyond depths of one to two meters. At the same time, blue–green components are transmitted more effectively [23]. Suspended particulates also scatter photons in both forward and backward directions, producing haze, edge blurring, and a characteristic low-contrast appearance [24]. Together, wavelength-selective absorption, volumetric scattering, and non-uniform artificial illumination define the underwater imaging degradation problem that robust visual detection pipelines must address.

Early computational approaches to underwater degradation relied on physical formation models. The Dark Channel Prior (DCP), originally proposed for single-image haze removal, provides an important prior-based foundation for visibility restoration [26]. Underwater adaptations and related single-image restoration methods, such as the method of Drews et al. [27], further estimated underwater depth and restored images under simplified imaging assumptions. However, such model-based methods often fail under the complex multi-scattering conditions of turbid water. Deep learning methods subsequently improved the quality of enhancement. Li et al. established the UIEB benchmark and proposed Water-Net as a learning-based baseline for underwater image enhancement [16]. Li, Anwar, and Porikli proposed UWCNN, a deep underwater image and video enhancement model inspired by underwater scene priors and trained on synthesized underwater

degradation [17]. Islam et al. introduced FUnIE-GAN, a fast GAN-based underwater image enhancement model designed for real-time visual perception in underwater robotic systems [18]. More recently, diffusion-based methods have advanced the quality frontier, including a diffusion-with-adversarial-learning framework published in the MDPI Journal of Imaging that performs strongly across multiple degradation types [19].

Transformer-based and state-space model-based enhancement architectures have recently become prominent. FACMamba proposed a Mamba-based frequency-aware enhancement framework with multi-directional spatial context modeling, published in the Journal of Marine Science and Engineering in 2025 [28]. Lin et al. introduced Dual-CycleGANs with dynamic guidance for robust underwater image restoration, also published in J. Mar. Sci. Eng. in 2025 [29]. A comparative study by Ma et al. systematically evaluated traditional and deep learning approaches for underwater remote sensing image enhancement across quantitative metrics, providing a useful benchmark for assessing trade-offs among enhancement paradigms [30]. Summers et al. further highlighted an underappreciated issue: image enhancement methods optimized for perceptual quality do not necessarily improve downstream vision tasks such as feature matching and navigation, underscoring the mismatch between enhancement and detection objectives [20].

This mismatch between enhancement and detection motivates AquaYOLO26. Rather than performing enhancement first and detection second, we integrate degradation compensation directly into the YOLO26 backbone through the proposed UWBN module. UWBN operates inside the detection pipeline and is optimized jointly with the detection objective.

2.2. YOLO-Based Object Detection: Evolution and Domain Adaptation

Since its introduction by Redmon et al. in 2016 [31], the YOLO framework has evolved through more than a dozen major releases. Key milestones include the Feature Pyramid Network (FPN) neck in YOLOv5 for multi-scale feature aggregation, the anchor-free detection head in YOLOv8 [32], programmable gradient information in YOLOv9, and attention-augmented feature reuse in YOLO11. Each generation has improved the accuracy–latency trade-off while expanding hardware compatibility. Sapkota et al. provide a comprehensive overview of this trajectory from YOLOv5 through YOLO26 [22].

The Ultralytics YOLO26 family introduces several architectural decisions that distinguish it from prior versions [21]. First, it adopts a native NMS-free, end-to-end inference architecture that removes the Non-Maximum Suppression post-processing step, reducing CPU latency by up to 43% in the nano variant compared with NMS-dependent predecessors. Second, Small Target-Aware Label Assignment (STAL) improves detection of objects with small spatial footprints by reweighting matching scores according to target scale. Third, the MuSGD optimizer, a hybrid of SGD and the Muon optimizer derived from advances in large language model training, improves training stability and convergence. Fourth, Progressive Loss Balancing (ProgLoss) dynamically modulates loss contributions over training, reducing early-stage instability on heterogeneous data distributions. Removal of Distribution Focal Loss (DFL) further simplifies the detection head, reducing parameter count while preserving localization quality.

YOLO has also proved adaptable to specialized domains beyond standard benchmarks. Cheng et al. showed that a lightweight YOLOv8s variant could accurately grade strawberry maturity after domain-specific architectural tuning, illustrating YOLO's versatility in precision agriculture [33]. In maritime applications, YOLO-SEA extended YOLOv8 by incorporating attention fusion and Soft-NMS for robust multi-class maritime target detection in adverse sea states [34]. For cross-domain detection, Zhang et al. proposed DA-YOLO, which integrates image-level and instance-level feature alignment into a one-stage YOLO framework to reduce domain shift [35]. YOLO-G further showed that adversarial cross-

domain feature alignment with gradient reversal can be embedded into a YOLO backbone with limited computational overhead [36]. The Hierarchical Multi-scale Domain Adaptive YOLO (HMDA–YOLO) framework proposed hierarchical backbone and head adaptation for more comprehensive distribution alignment in one-stage cross-domain detection [37]. These domain-adaptive YOLO methods provide the conceptual basis for DA-ProgLoss, which adapts gradient reversal to the progressive loss-scheduling framework of YOLO26.

2.3. Underwater Object Detection

Underwater object detection inherits the standard challenges of object detection—occlusion, scale variation, and cluttered backgrounds—while also facing the underwater degradation effects described in Section 2.1. The distinctive visual properties of aquatic environments have motivated a growing body of YOLO-based underwater detection architectures.

SEANet introduced a Semantic Enhancement and Amplification Network that combines multi-scale detail amplification with foreground–background contrast attention to address low-contrast, complex-background underwater scenes [9]. AGS-YOLO proposed a multi-scale attention module for small underwater target detection under low-resource constraints and combined it with a streamlined GSConv structure to reduce parameter count [11]. MD-YOLO introduced deformable attention in the detection neck to improve localization of debris targets with imbalanced aspect ratios [12]. IFEM-YOLOv13 introduced feature-symmetry balancing to address image degradation, low contrast, and small target obscurity under underwater conditions, explicitly linking symmetric architectural principles to detection robustness [13]. UDD-YOLO integrated a Cold Diffusion model as a preprocessing stage within a YOLO detection framework, applying deterministic degradation reversal to improve underwater image clarity [14].

Earlier foundational contributions include YOLO-GE, which introduced attention-based feature fusion for underwater target detection [10], and YOLOv8-MU, which incorporated a large-kernel block and multi-branch reparameterization to improve underwater receptive field modeling [38]. An energy-efficient adaptive vision transformer for underwater detection further explored adversarial robustness and energy efficiency as joint objectives for embedded underwater systems [39]. More recently, an improved YOLOv11 framework for marine debris detection, incorporating MixStructureBlock and Efficient Multi-scale Attention (EMA), achieved state-of-the-art mAP@0.5 on TrashCan-Instance [40]. A sonar-oriented AquaYOLO variant was also proposed for underwater acoustic imaging [41].

Existing methods can be broadly grouped into three paradigms: (1) two-stage enhance-then-detect pipelines that apply image enhancement as preprocessing; (2) attention or module augmentation of standard YOLO backbones to improve feature robustness; and (3) joint or end-to-end frameworks that couple restoration with detection. EnYOLO is an important example of paradigm integrating underwater image enhancement (UIE) and underwater object detection (UOD) within a shared backbone and using multi-stage training to balance the two tasks [25]. AquaYOLO26 differs from all three paradigms: rather than relying on explicit image enhancement, module stacking, or dual-task joint training, it embeds physical degradation priors directly into the normalization layers of the YOLO26 backbone, enabling detection-oriented compensation without the overhead or objective mismatch of conventional approaches.

2.4. Underwater Marine Debris Datasets and Domain Diversity

The availability of annotated underwater debris datasets remains a significant bottleneck in this research area, and dataset characteristics directly shape experimental design. This work uses three datasets, each representing a distinct distribution within the broader spectrum of underwater imaging conditions.

The SeaClear Marine Debris Dataset [6], published in Scientific Data in 2024, serves as the primary training dataset in this study. It contains 8610 annotated ROV images collected across multiple shallow-water coastal regions in Croatia and France, covers 40 target categories, and supports both object detection and instance segmentation. Its multi-site, multi-camera collection protocol embeds domain shift directly into the dataset structure, making it well-suited for studying cross-location generalization—a property emphasized by the original authors. SeaClear captures shallow-water, natural light coastal conditions with diverse debris appearances against organic seafloor backgrounds.

TrashCan 1.0 [7] provides approximately 7212 annotated images from deep-water ROV footage, covering debris, ROV structural components, and marine organisms. Its deep-water context, historical ROV imagery, and complex biological background elements produce a distribution that differs substantially from SeaClear, with lower ambient light, different color statistics, and more frequent co-occurrence of debris with marine fauna.

Trash-ICRA19 [8] originates from underwater robotics research and consists of video-extracted frames annotated with bounding boxes across debris, biological, and ROV categories. Its relatively small scale and distinct frame extraction methodology, compared with SeaClear and TrashCan, make it suitable as a held-out zero-shot generalization benchmark: the dataset is never used during training, and performance on it directly measures robustness to cross-dataset transfer.

Together, these datasets span shallow coastal, deep-water, and robotic survey imaging conditions, enabling a more comprehensive evaluation of cross-domain generalization than single-dataset reporting. This multi-dataset protocol is designed to reveal models that overfit to the statistics of one domain while appearing competitive on within-domain metrics. The cross-domain gaps among the datasets primarily arise from differences in color statistics, turbidity, camera response curves, and debris appearance frequency distributions.

2.5. Domain Adaptation and Cross-Domain Generalization in Object Detection

Domain shift—the statistical mismatch between training and deployment data distributions—is a fundamental challenge in applied machine learning. In underwater contexts, this challenge is especially severe because optical conditions vary substantially across water bodies, depths, and geographic regions [25]. Domain adaptation methods reduce this mismatch by aligning source and target feature representations. In contrast, domain generalization methods aim to learn representations that are invariant to domain-specific statistics without access to target domain data during training.

The domain-adversarial neural network framework proposed by Ganin et al. [42] establishes the core principle of adversarial domain adaptation: a gradient reversal layer inserted between the feature extractor and a domain classifier produces domain-invariant representations by reversing the gradients of the domain classification loss during back-propagation. This principle has been widely applied to object detection via frameworks such as DA-YOLO [35] and YOLO-G [36], which demonstrate that single-stage YOLO detectors can achieve meaningful cross-domain improvements through adversarial feature alignment with minimal architectural modifications.

Recent work has further integrated domain adaptation into training-loss schedules. HMDA–YOLO proposed hierarchical multi-scale domain alignment across both the backbone and detection head, showing that comprehensive distribution alignment, rather than alignment at a single network stage, yields more stable cross-domain performance [37]. Style transfer methods provide a complementary route by translating image appearance across domains and reducing low-level input shift. StyDeco, which uses distilled priors and semantic decoupling for unsupervised style transfer [43], illustrates how disentangling style and content representations can enable cross-domain visual alignment without

paired data—a methodological principle reflected in the domain alignment branch of DA-ProgLoss.

Within underwater detection, WQT + DG-YOLO combines water quality-based data augmentation with domain-generalized YOLO training to improve robustness under varying water quality conditions [44]. The domain similarity-perceived label assignment approach [45] further shows that adapting label assignment to source–target domain similarity can improve cross-domain small target detection. This idea is conceptually related to Turb-STAL, which adapts YOLO26's STAL according to per-image turbidity estimates rather than explicit domain labels.

AquaYOLO26 synthesizes these directions through DA-ProgLoss: it uses YOLO26's ProgLoss schedule as the optimization backbone, adds a gradient reversal domain-adversarial branch operating across source (SeaClear) and target (TrashCan 1.0) domains during training, and evaluates generalization through zero-shot transfer to Trash-ICRA19. This design yields a rigorous three-dataset protocol for assessing cross-water-domain robustness.

3. Methodology

In this study, underwater degradation is interpreted as a feature asymmetry problem rather than merely an image quality degradation problem. Accordingly, AquaYOLO26 does not pursue perceptual image restoration; instead, it aims to restore the balance of detection-relevant features under channel-asymmetric attenuation, turbidity-induced small target suppression, and cross-domain distribution asymmetry.

This section presents the complete design of AquaYOLO26. We first describe the overall architecture and its relationship to the YOLO26 baseline (Section 3.1). We then detail the three proposed contributions: Underwater-aware Batch Normalization (Section 3.2), Turbidity-conditioned STAL (Section 3.3), and Domain-Adversarial ProgLoss (Section 3.4). Section 3.5 summarizes the full training configuration.

3.1. Overall Architecture

AquaYOLO26 is designed around a symmetry restoration principle for degraded underwater perception. In clear conditions, object detectors rely on coherent, well-balanced feature responses. Underwater environments break this balance. To rigorously ground our methodology, we formally define feature symmetry in this study as the representational and statistical parity across three specific dimensions, and we design corresponding modules to measure and restore them:

1. Channel Symmetry (Spectral Parity): The statistical balance of feature responses across wavelength-related channel groups. Wavelength-dependent light absorption asymmetrically suppresses red channel signals. We measure the restoration of channel symmetry by computing the spectral symmetry index (S_{spec}) from actual internal feature activation magnitudes across channel groups within the backbone normalization layers. UWBN is designed to promote zero mean and unit variance statistics across the compensated feature channels, thereby restoring empirical statistical parity among the channel groups during feature extraction.
2. Scale Symmetry (Spatial Parity): The balance in detection capability between macroscopic and microscopic objects. Turbidity-induced scattering asymmetrically degrades high-frequency details, suppressing small targets. We measure this symmetry using Small Target mAP (mAP_{small}) and Small Target Recall (SR@0.5). Turb-STAL (Section 3.3) restores this parity by adaptively reweighting assignment scores for obscured small instances.
3. Domain Symmetry (Distributional Parity): The alignment of feature distributions across heterogeneous water environments. Severe domain shifts create asymmetrical

generalization capabilities (high source accuracy, low target accuracy). We measure domain symmetry using the Domain Invariance Ratio ($\rho = \mathrm{mAP}_{\mathrm{target}}/\mathrm{mAP}_{\mathrm{source}}$). DA-ProgLoss (Section 3.4) restores this symmetry via adversarial alignment, pushing ρ closer to the ideal symmetric state of 1.0.

AquaYOLO26 is built on the YOLO26 baseline [21], which follows a standard three-stage detector structure: a convolutional backbone for hierarchical feature extraction, a Path Aggregation Network (PAN) neck for multi-scale feature fusion, and an anchor-free, NMS-free detection head. YOLO26 removes the Distribution Focal Loss (DFL) head and replaces post-processing Non-Maximum Suppression with a native end-to-end decoding pipeline, reducing CPU-side latency by up to 43% compared with prior YOLO variants [21,22].

AquaYOLO26 introduces modifications at three stages of the pipeline without changing the neck or head architecture, thereby preserving YOLO26's NMS-free inference advantages and full hardware compatibility:

1. Backbone (normalization layers): Each standard Batch Normalization (BN) layer in the YOLO26 backbone is replaced by UWBN, which embeds physics-guided wavelength attenuation compensation into normalization. This is the only structural modification to the backbone computation graph.
2. Label assignment (training only): YOLO26's STAL is replaced by Turb-STAL, which incorporates a lightweight per-image turbidity index into the assignment score and dynamic anchor candidate count. Turb-STAL operates only during training; at inference, the standard task-aligned prediction head remains unchanged.
3. Training loss and domain alignment: YOLO26's ProgLoss is extended to DA-ProgLoss by attaching a gradient reversal domain classifier to the PAN-neck feature representation during training. This branch aligns source domain SeaClear features with target domain TrashCan 1.0 features, encouraging cross-water-domain invariance. The domain-adversarial branch is removed at inference and therefore introduces no inference latency.

As shown in Figure 1, AquaYOLO26 preserves the overall YOLO26 detection pipeline while introducing underwater-specific modifications in backbone normalization, training-time label assignment, and domain-adversarial loss branches. Turb-STAL and DA-ProgLoss are used only during training and add no inference time computation, whereas only UWBN remains in the deployed computation graph, increasing the parameter count from 2.42 M to 2.46 M with only a marginal impact on inference latency, as quantified in Section 5.5.

3.2. Underwater-Aware Batch Normalization (UWBN)

3.2.1. Physical Motivation of UWBN

Light traveling through water follows exponential attenuation governed by the Beer–Lambert law [23]:

$$I_c(d) = I_0 \cdot e^{-\alpha_c \cdot d} \tag{1}$$

where $I_c(d)$ is the irradiance at wavelength channel c after propagating to depth d, I_0 is the surface irradiance, and α_c is the wavelength-dependent attenuation coefficient. For clear water, $\alpha_{red} \approx 0.80\ \mathrm{m}^{-1}$, $\alpha_{green} \approx 0.04\ \mathrm{m}^{-1}$, and $\alpha_{\mathrm{blue}} \approx 0.01\ \mathrm{m}^{-1}$, producing the well-known rapid decay of the red signal with depth [24].

Standard Batch Normalization, applied uniformly across feature channels, is insensitive to channel-asymmetric signal loss. BN therefore normalizes degraded and undegraded channels identically, allowing attenuation patterns to persist as systematic bias throughout the feature hierarchy. UWBN addresses this issue by estimating effective imaging depth from input feature statistics and applying channel-wise precompensation before standard normalization.

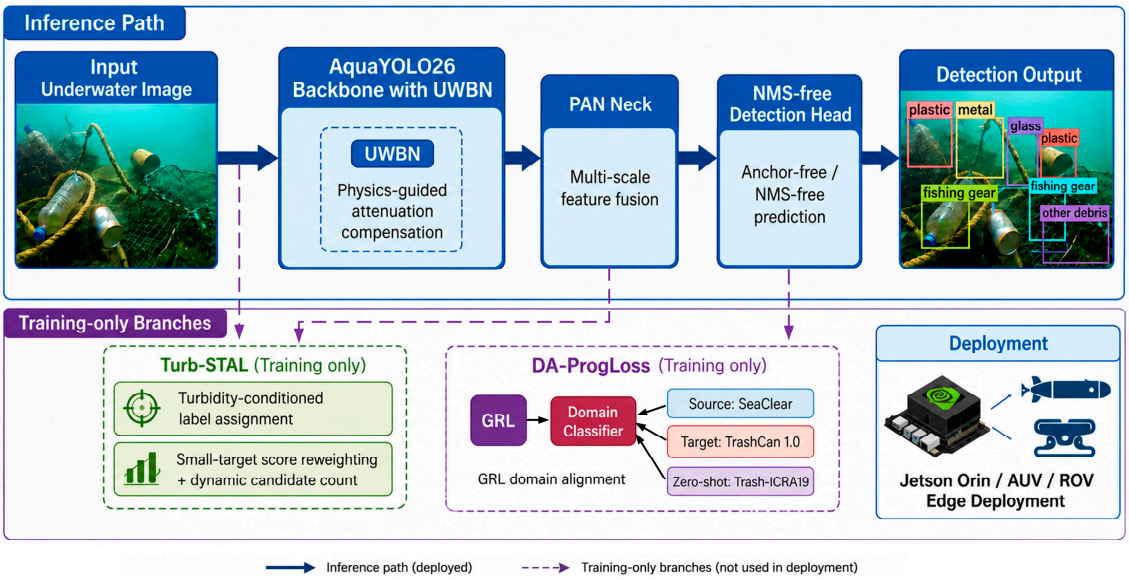


Figure 1. AquaYOLO26 pipeline. GRL denotes the Gradient Reversal Layer used in the DA-ProgLoss branch (Section 3.4.3). The proposed framework is built on the YOLO26 architecture with three key modifications: UWBN replaces standard batch normalization in the backbone to provide degradation-aware feature compensation; Turb-STAL is applied during training to adapt label assignment under varying turbidity conditions; and the DA-ProgLoss branch performs domain-adversarial feature alignment through the gradient reversal layer. These modules are integrated into a unified training framework, whereas only the backbone modification, UWBN, is retained during deployment, thereby preserving the advantages of YOLO26's NMS-free inference pipeline. The additional inference time parameters introduced by UWBN, reported as 0.04 M and corresponding to approximately 1.65% of the 2.42 M-parameter YOLO26n baseline, have only a marginal impact on inference latency. Different colors and arrows indicate different functional components and information flows. Dashed boxes denote training-only modules.

3.2.2. Depth Proxy Estimation

At each backbone layer, UWBN estimates a soft depth proxy $\hat{d} \in \mathbb{R}^+$ from the current feature map $X \in R^{B\times C\times H\times W}$. For the input layer $(C = 3)$, the three channels correspond directly to the physical $\{R, G, B\}$ color channels. For deeper backbone layers $(C > 3)$, the C channels are partitioned into three contiguous, approximately equal-sized channel groups G_1, G_2, G_3 along the channel dimension:

$$G_1 = \{1, \ldots, C_g\}, G_2 = \{C_g + 1, \ldots, 2C_g\}, G_3 = \{2C_g + 1, \ldots, C\} \quad (2)$$

where $C_g = \lfloor C/3 \rfloor$ denotes the subgroup channel capacity (with any remaining channels assigned to G_3).

The group mean magnitude $\bar{x}_{G_k}$ for each group $k \in \{1, 2, 3\}$ is computed as:

$$\bar{x}_{G_k} = \frac{1}{|G_k| \cdot H \cdot W} \sum_{c\in G_k} \sum_{h=1}^{H} \sum_{w=1}^{W} X_{b,c,h,w} \quad (3)$$

The depth proxy $\hat{d}$ is then computed as:

$$\hat{d} = \text{softplus}\left(-\frac{1}{3} \sum_{k=1}^{3} \frac{\bar{x}_{G_k}}{x_{ref}} \right) \quad (4)$$

where x_{ref} is a running exponential moving average of the mean feature magnitude maintained across training batches to stabilize feature scale across iterations. The softplus activation guarantees $\hat{d} > 0$.

This channel partitioning is fixed and deterministic and does not introduce auxiliary grouping parameters. Because intermediate feature channels no longer retain literal RGB semantics after convolutional transformations, the three-group mapping should be interpreted as a physics-inspired structural approximation rather than as a direct correspondence between deep feature channels and physical color channels. The adaptive attenuation responses are learned through the channel-wise parameters in Equation (6).

3.2.3. Attenuation Compensation and Normalization

Given the estimated depth proxy, UWBN first applies channel-wise exponential precompensation:

$$\widetilde{x}_{b,c,h,w} = x_{b,c,h,w} \cdot e^{\hat{\alpha}_c \cdot \hat{d}} \tag{5}$$

where $\hat{\alpha}_c$ is a learnable scalar per feature channel initialized via a linear projection from the three physical RGB absorption coefficients $\alpha_{phys} = [0.80, 0.04, 0.01]^\top$:

$$\hat{\alpha} = W_\alpha \alpha_{\text{phys}} + b_\alpha,\ W_\alpha \in \mathbb{R}^{C\times 3}, b_\alpha \in \mathbb{R}^C \tag{6}$$

The parameters $\{W_\alpha, b_\alpha\}$ are learned jointly with the backbone via gradient descent; their physics-based initialization provides a physically motivated inductive bias for optimization.

UWBN then applies standard normalization to the compensated features:

$$\begin{aligned} \hat{x}^{UW}_{b,c,h,w} &= \frac{\widetilde{x}_{b,c,h,w} - \widetilde{\mu}_c}{\sqrt{\widetilde{\sigma}^2_c + \epsilon}} \\ y_{b,c,h,w} &= \gamma_c \cdot \hat{x}^{UW}_{b,c,h,w} + \beta_c \end{aligned} \tag{7}$$

where $\widetilde{\mu}_c$ and $\widetilde{\sigma}^2_c$ are the batch mean and variance of the compensated features; $\widetilde{x}_c$, and $\{\gamma_c, \beta_c\}$ are the standard learnable affine parameters of BN. The complete per-layer learnable parameter set of UWBN is $\{W_\alpha, b_\alpha, \gamma_c, \beta_c\}$.

Design Rationale for Operation Order: Attenuation compensation (Equation (5)) should precede batch normalization (Equation (7)). If standard BN were applied first, it would independently shift and scale each feature channel to a zero mean, unit variance distribution, effectively destroying the relative magnitude differences across channels that characterize the physical attenuation signature. Furthermore, applying exponential compensation to post-BN features (which can include negative values due to mean subtraction) is physically inconsistent and can cause numerical instability. By compensating first, UWBN physically restores the degraded signal magnitudes, ensuring that the subsequent normalization step acts upon a symmetric and balanced feature space before passing it to the next convolutional layer. In preliminary experiments, reversing the operation order resulted in less stable optimization and lower mAP@0.5. Because this preliminary comparison was not included in the formal repeated run evaluation, we report it only as an implementation observation rather than as a formal quantitative conclusion.

The critical distinction from a naive UIE preprocessing approach is that Equations (5)–(7) are differentiable with respect to all parameters and are trained end-to-end with the detection loss. Thus, $\hat{\alpha}_c$ adapts to maximize detection performance rather than perceptual image quality.

Figure 2 illustrates the computational flow of UWBN: the input feature map is used to estimate a depth proxy, channel-wise attenuation correction is then applied, and standard batch normalization is finally performed on the compensated features.

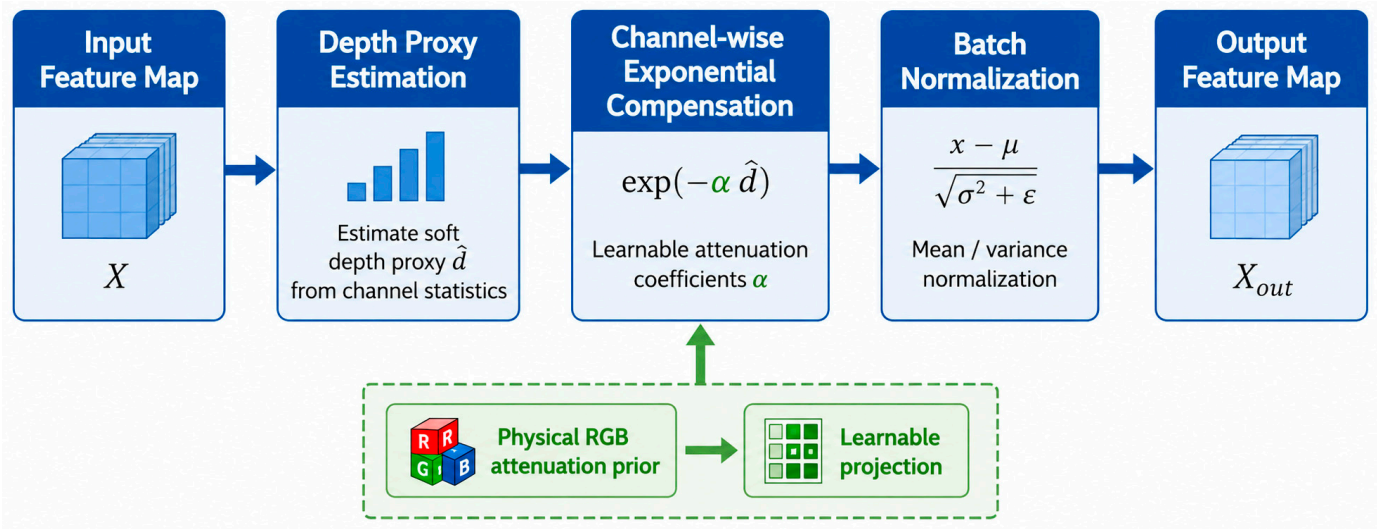


Figure 2. UWBN module. Input feature map X → depth proxy estimation ($\hat{d}$, Equation (4)) → channel-wise exponential compensation ($\tilde{X}$, Equation (5)) → batch normalization (Equation (7)) → output Y. The dashed box shows the learnable attenuation coefficient projection $\{W_\alpha, b_\alpha\}$ (Equation (6)).

Physical Justification and Constraints

The physical prior initialization for the attenuation coefficient α_c ensures that the framework remains grounded in oceanographic reality:

Spectral Mapping: For feature maps with C channels, we partition the capacity into three contiguous groups G_1, G_2, G_3. Each group is assigned a corresponding initial attenuation coefficient $\alpha_{c,init} \in \{0.80, 0.04, 0.01\}$ based on the Beer–Lambert law. This initialization provides a physics-inspired inductive bias for channel-wise compensation, but it does not imply that intermediate feature channels retain literal red, green, or blue identities.

Positivity and Hierarchy Constraints: To prevent the learned coefficients from becoming physically nonsensical (e.g., negative attenuation), the learnable parameters in Equation (6) are initialized with the Beer–Lambert coefficients and optimized via a softplus boundary, ensuring $\alpha_c > 0$. The initialization encodes the expected spectral attenuation hierarchy, while the softplus parameterization guarantees that the learned coefficients remain positive. Because these coefficients operate on generalized feature channel groups, they should not be interpreted as direct wavelength-specific attenuation measurements.

Environment-Conditioned Effective Compensation: The learned channel-wise attenuation coefficients are global model parameters and remain fixed after training. In contrast, the image-conditioned depth proxy varies across underwater conditions. Consequently, the effective compensation magnitude, determined jointly by the learned attenuation coefficient and the estimated depth proxy, tends to be larger for images with stronger estimated degradation. This variation should be interpreted as image-dependent effective compensation rather than as an image-dependent change in the learned attenuation coefficients themselves.

3.3. Turbidity-Conditioned STAL (Turb-STAL)

3.3.1. Motivation of Turb-STAL

YOLO26's Small Target-Aware Label Assignment (STAL) extends the task-aligned assignment framework by augmenting the alignment score and the top-k candidate set to improve matching quality for small objects [21]. The standard STAL alignment score for an

anchor candidate a with predicted class distribution $\hat{p}(a)$ and box prediction $\hat{b}_a$, matched against ground truth instance g with class label c_g and box b_g, is:

$$s_{STAL}(a, g) = \hat{p}_{c_g}(a)^{\alpha_{cls}} \cdot IoU(\hat{b}_a, b_g)^{\alpha_{iou}} \tag{8}$$

where α_{cls} and α_{iou} are hyperparameters balancing classification and localization terms.

Under high turbidity, suspended particles scatter the light signal of small debris targets, suppressing the confidence $\hat{p}_{c_g}(a)$ of candidate anchors near the true target location and reducing their computed IoU relative to clear-water conditions. As a result, STAL may systematically underrepresent small debris in high-scattering images, assigning their ground truth instances to fewer, lower-quality candidates. Turb-STAL addresses this by conditioning both the alignment score and the candidate count on a per-image turbidity estimate.

3.3.2. Turbidity Index Estimation

A lightweight turbidity estimator T is attached to the last stage backbone feature map $F_{back} \in \mathbb{R}^{B \times C_{back} \times H_b \times W_b}$:

$$\begin{aligned} \mathrm{f_{gap}} &= \mathrm{GlobalAvgPool}(\mathrm{F_{back}}) \in \mathbb{R}^{\mathrm{B} \times \mathrm{C_{back}}} \\ \tau &= \sigma\left(\mathrm{w_2^\top} \cdot \mathrm{ReLU}\left(\mathrm{w_1^\top f_{gap}} + \mathrm{b_1}\right) + \mathrm{b_2}\right) \end{aligned} \tag{9}$$

where $w_1 \in \mathbb{R}^{C_{back} \times 64}, b_1 \in \mathbb{R}^{64}$, $w_2 \in \mathbb{R}^{64}, b_2 \in \mathbb{R}$, and σ is the sigmoid function. The output $\tau \in [0, 1]$ represents the estimated turbidity index for the current image, with $\tau = 0$ indicating clear water and $\tau = 1$ indicating maximal scattering.

The turbidity estimator is trained without explicit turbidity supervision by using a proxy objective. Specifically, motivated by wavelength-dependent underwater attenuation [23,24], we use the per-channel red-to-blue mean intensity ratio as a weak proxy target:

$$\mathrm{L_{turb}} = \left(\tau - \sigma\left(\beta_{\mathrm{proxy}} \cdot \left(1 - \frac{\bar{\mathrm{I}}_{\mathrm{R}}}{\bar{\mathrm{I}}_{\mathrm{B}} + \epsilon}\right)\right)\right)^2 \tag{10}$$

where $\bar{I}_R$ and $\bar{I}_B$ are the mean red and blue channel intensities of the input image, and $\beta_{proxy} = 5.0$ is a scaling factor. This auxiliary loss is weighted at 0.01 relative to the detection loss.

A fixed red-to-blue intensity ratio is a useful, albeit weak, indicator of underwater degradation. Still, it is less suitable as the sole turbidity signal for directly controlling label assignment. Because the ratio is computed from global RGB intensity statistics, it may conflate turbidity-related scattering with spectral variations caused by illumination, depth-dependent attenuation, or seabed reflectance. In addition, such a handcrafted index is not conditioned on the feature representation used for detection and label assignment. We therefore use the red-to-blue ratio only as a proxy target and predict the image-level turbidity index from the last stage backbone feature map. This design provides Turb-STAL with a feature-conditioned control signal without directly applying a fixed raw image index to the assignment process.

Nevertheless, the learned estimator remains affected by the limitations of its proxy supervision. Artificial illumination with atypical spectral characteristics may alter the red-to-blue balance without a corresponding change in the severity of scattering. Similarly, in very shallow water, sun glint, surface reflections, and bottom-reflected light may contaminate the proxy signal. Moreover, because the current estimator outputs a single image-level index, it cannot explicitly represent spatially non-uniform turbidity caused by local sediment plumes or uneven illumination. These limitations may lead to overestima-

tion or underestimation of turbidity, and their potential implications for label assignment and detection are discussed further in Section 5.6.

To assess whether the learned turbidity index captures meaningful underwater degradation variation, we examined its association with an external image quality metric and manually assigned degradation categories.

1. UCIQE Correlation: We computed the UCIQE (Underwater Color Image Quality Evaluation) metric—a recognized proxy for visibility and contrast—for all images. Our analysis revealed a strong negative correlation (Pearson coefficient $r = -0.78$, $p < 0.001$) between τ and UCIQE. This indicates that as image quality degrades (lower UCIQE), the turbidity estimator consistently predicts a higher τ value, supporting the interpretation that the proxy is sensitive to visibility loss.
2. Categorical Separation: We manually classified 500 images into three degradation categories (Clear, Moderate, Severe) based on the observable scattering intensity. The turbidity estimator yielded mean τ values of 0.13 ± 0.04 for Clear images, 0.52 ± 0.09 for Moderate images, and 0.89 ± 0.05 for Severe images. A one-way ANOVA showed a significant difference among the three categories ($F(2, 497) = 412.85$, $p < 0.001$), indicating that the estimated index differs systematically across the three manually assigned degradation levels.

These analyses support interpreting τ as a degradation-related proxy. However, τ should not be interpreted as a direct physical measurement of turbidity concentration.

3.3.3. Modified Alignment Score and Candidate Count

Turb-STAL modifies STAL in two ways. First, for each ground truth instance g with relative bounding box area $A_g \in (0, 1]$ (normalized by image area), a turbidity-adaptive weight amplifies the alignment score of small targets:

$$w_{turb}(g, \tau) = 1 + \lambda_\tau \cdot \tau \cdot 1[A_g < \delta_s] \tag{11}$$

where $\lambda_\tau = 0.5$ and $\delta_s = 0.01$ (targets occupying less than 1% of the image area are considered small-scale). The modified alignment score is:

$$s_{TurbSTAL}(a, g) = s_{STAL}(a, g) \cdot w_{turb}(g, \tau) \tag{12}$$

Second, the number of top-k candidate anchors retained per ground truth instance is made turbidity-adaptive:

$$k_{turb}(\tau) = \lfloor k_0 \cdot (1 + \beta_k \cdot \tau) \rfloor \tag{13}$$

where $k_0 = 10$ is the base YOLO26 STAL candidate count and $\beta_k = 1.0$, yielding $k_{turb} \in [10, 20]$. Under high turbidity ($\tau \to 1$), the candidate pool doubles, compensating for the degraded alignment scores by providing additional assignment opportunities for each ground truth instance.

The final label assignment for each ground truth instance g selects the top-$k_{turb}(\tau)$ candidates ranked by $s_{TurbSTAL}$, with conflict resolution (anchors assigned to multiple ground truths) resolved by highest score priority, consistent with the original YOLO26 STAL protocol.

Figure 3 summarizes the Turb-STAL assignment procedure, showing how the estimated turbidity index jointly controls small target score amplification and the adaptive candidate-anchor count.

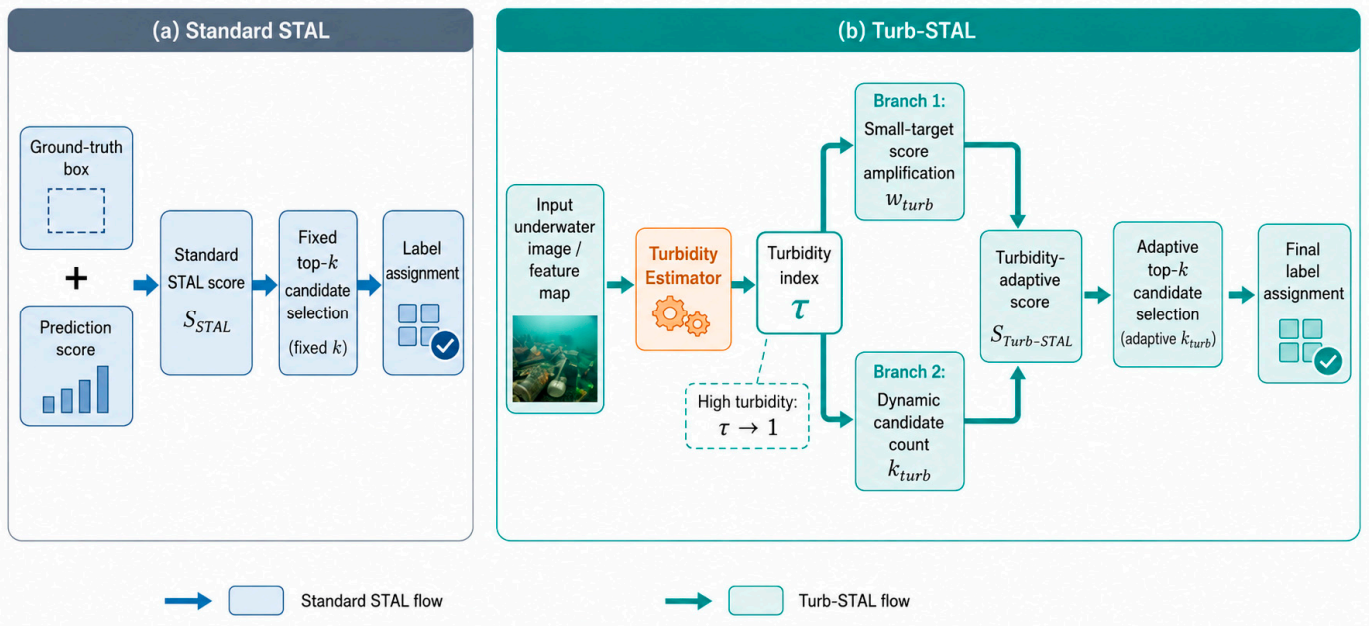


Figure 3. Turb-STAL label assignment. (**a**) standard STAL candidate selection with fixed $k = k_0$ and score s_{STAL}. (**b**) Turb-STAL uses the turbidity index τ estimated by the turbidity estimator (Equation (9)) to condition both the small target score amplification through w_{turb} (Equation (11)) and the dynamic candidate count k_{turb} (Equation (13)). Under high turbidity ($\tau \approx 1$), small target candidates receive amplified scores and an expanded candidate pool, recovering assignment quality lost to feature suppression.

3.4. Domain-Adversarial ProgLoss (DA-ProgLoss)

3.4.1. Motivation of DA-ProgLoss

Despite strong within-domain performance, detectors trained on a single underwater dataset often generalize poorly to unseen underwater environments because color statistics, turbidity profiles, and background composition vary substantially across domains [25]. YOLO26's ProgLoss progressively rebalances detection loss components over the training schedule [21], but it does not address the source-to-target distribution gap that limits cross-dataset transfer. DA-ProgLoss extends this progressive framework with an adversarial domain alignment branch, enabling the backbone to learn domain-invariant features while the detection head learns discriminative, class-specific representations. In practical robotic marine operations, unsupervised domain adaptation (UDA) based on target domain visual features is a realistic and cost-effective strategy. Before physical collection, AUVs and ROVs typically perform a rapid, non-contact reconnaissance survey of a target debris site. This phase produces abundant target domain video, whereas expert annotation in remote field environments is slow and expensive. By using 1000 unlabeled target domain frames (Section 4.1.2) for DA-ProgLoss alignment, the onboard edge computing unit can adapt the feature extractor to the optical profile of the new water environment during local adaptation cycles. This self-adaptive pipeline avoids human-in-the-loop annotation and is more practical than semi-supervised fine-tuning under real maritime constraints.

3.4.2. YOLO26 ProgLoss Baseline

Let the detection loss at training step t be:

$$L_{det}(t) = w_{box}(t) \cdot L_{box} + w_{cls}(t) \cdot L_{cls} + w_{obj}(t) \cdot L_{obj} \quad (14)$$

where L_{box} is the Complete IoU (CIoU) bounding box regression loss, L_{cls} is the binary cross-entropy classification loss, L_{obj} is the objectness confidence loss, and $w_{box}(t)$, $w_{cls}(t)$,

$w_{obj}(t)$ are time-varying progressive weights. The ProgLoss schedule transitions each weight smoothly from an initial value $w_i^{(0)}$ to a final value $w_i^{(T)}$ over T total training steps:

$$\mathrm{w_i(t)} = \mathrm{w_i^{(0)}} + \left(\mathrm{w_i^{(T)}} - \mathrm{w_i^{(0)}}\right) \cdot \frac{1 - \cos(\pi \cdot \mathrm{t/T})}{2} \tag{15}$$

This follows a cosine annealing schedule. The default YOLO26 settings are $\left\{w_{box}^{(0)}, w_{box}^{(T)}\right\} = \{7.5, 7.5\}$, $\left\{w_{cls}^{(0)}, w_{cls}^{(T)}\right\} = \{0.5, 0.5\}$, $\left\{w_{obj}^{(0)}, w_{obj}^{(T)}\right\} = \{1.5, 1.5\}$, with minor adjustments for the multi-scale output head.

3.4.3. Gradient Reversal Layer and Domain Classifier

Following the DANN framework of Ganin et al. [42], we attach a domain classifier $\mathrm{D_{cls}}$ to a bottleneck feature map $\mathrm{F_{neck}} \in \mathbb{R}^{\mathrm{B}\times \mathrm{C_{neck}} \times \mathrm{H_n} \times \mathrm{W_n}}$ extracted from the last stage of the PAN neck via a Gradient Reversal Layer (GRL):

$$\begin{aligned} \widetilde{\mathrm{F}}_{\mathrm{neck}} &= \mathrm{GRL}_{\lambda_{\mathrm{DA}}(\mathrm{t})}(\mathrm{F_{neck}}) \\ \hat{\mathrm{d}} &= \mathrm{D_{cls}}(\mathrm{GlobalAvgPool}(\widetilde{\mathrm{F}}_{\mathrm{neck}})) \end{aligned} \tag{16}$$

where D_{cls} is a three-layer fully connected network with architecture $[C_{neck} \to 512 \to 128 \to 1]$ with ReLU activations and a sigmoid output for binary source/target domain classification. The GRL reverses gradients flowing from the domain classifier back into the backbone during backpropagation:

$$\frac{\mathrm{d}\widetilde{\mathrm{F}}_{\mathrm{neck}}}{\mathrm{dF_{neck}}}\Big|_{\mathrm{forward}} = \mathrm{I}, \quad \frac{\mathrm{dL_{dom}}}{\mathrm{dF_{neck}}}\Big|_{\mathrm{backward}} = -\lambda_{\mathrm{DA}}(\mathrm{t}) \cdot \frac{\mathrm{dL_{dom}}}{\mathrm{d}\widetilde{\mathrm{F}}_{\mathrm{neck}}} \tag{17}$$

where L_{dom} is the domain classification loss:

$$L_{dom} = -\mathbb{E}_{x \sim D_s \cup D_t}\left[d \log \hat{d} + (1-d)\log\left(1-\hat{d}\right)\right]$$

with $d = 0$ for source domain images (SeaClear) and $d = 1$ for target domain images (TrashCan 1.0).

3.4.4. Progressive Domain Alignment Schedule

The reversal coefficient $\lambda_{\mathrm{DA}}(\mathrm{t})$ is set to increase progressively throughout training, following the standard DANN schedule of Ganin et al. [42]:

$$\lambda_{\mathrm{DA}}(\mathrm{t}) = \lambda_{\max} \cdot \frac{2}{1 + \exp(-\gamma \cdot \mathrm{t/T})} - 1 \tag{18}$$

With $\lambda_{max} = 1.0$ and $\gamma = 10$, this schedule keeps domain alignment pressure low at the beginning of training (allowing the detection objective to establish stable representations before domain confusion begins). It reaches maximum strength in the later training epochs.

3.4.5. Total DA-ProgLoss Objective

The full DA-ProgLoss training objective is:

$$\mathrm{L_{total}(t)} = \mathrm{L_{det}(t)} + \lambda_{\mathrm{turb}} \cdot \mathrm{L_{turb}(t)} + \lambda_{\mathrm{DA}}(\mathrm{t}) \cdot \mathrm{L_{dom}(t)} \tag{19}$$

where the negative sign before the domain-adversarial term reflects the adversarial objective; through the gradient reversal layer, the backbone is optimized to confuse the source and target domains by maximizing the domain classification loss. Conversely, L_{dom} the

domain classifier itself is optimized to minimize L_{dom} and to predict the domain label for each feature map correctly. The coefficient $\lambda_{turb} = 0.01$ weights the auxiliary loss for the turbidity proxy.

At inference, neither the turbidity estimator auxiliary branch nor the GRL domain classifier is retained in the computation graph. The inference model is architecturally identical to the YOLO26 baseline, except that UWBN replaces standard BN in the backbone.

Figure 4 shows the complete DA-ProgLoss training objective, including the standard detection branch, the gradient reversal domain classifier, and the progressive domain alignment schedule.

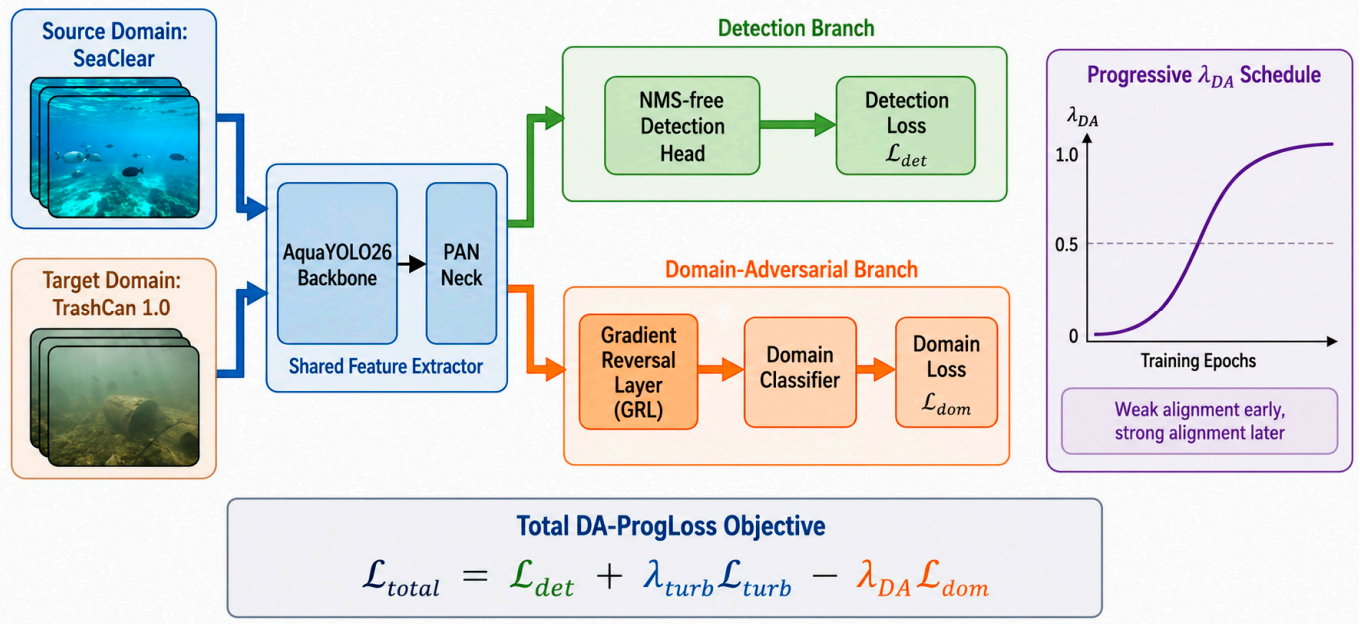


Figure 4. DA-ProgLoss training objective. The top branch shows the standard YOLO26 detection objective L_{det} (Equation (14)) and progressive loss weighting (Equation (15)). The lower branch shows the GRL-based domain-adversarial module attached to the PAN-neck output, including feature reversal and domain prediction (Equation (16)), gradient reversal (Equation (17)), and the progressive reversal schedule $\lambda_{DA(t)}$ (Equation (18)). The total training objective combines detection loss, turbidity auxiliary supervision, and adversarial domain alignment as shown in Equation (19). The domain classifier and turbidity auxiliary branch are used only during training.

3.5. Training Configuration

Baseline and backbone. AquaYOLO26 is initialized from official YOLO26n/s pretrained weights [21], pretrained on the COCO dataset [46]. UWBN weights $\{W_\alpha, b_\alpha\}$ are initialized as described in Section 3.2.3 and Physical Justification and Constraints; BN affine parameters $\{\gamma_c, \beta_c\}$ are loaded from the pretrained checkpoint.

Optimizer. We employ YOLO26's MuSGD hybrid optimizer [21] with an initial learning rate of $\eta_0 = 0.01$ and a cosine annealing schedule to a minimum of $\eta_{min} = 10^{-4}$. Momentum is set to 0.937 and weight decay to 5×10^{-4}.

Training schedule. Models are trained for 200 epochs with a batch size of 16 on a single NVIDIA RTX 4090 GPU. The first 3 epochs employ linear warm-up from $0.1\eta_0$ to η_0. For DA-ProgLoss training, SeaClear images and TrashCan 1.0 images are interleaved at a 1:1 ratio within each training batch to provide balanced domain supervision.

Input resolution. All images are resized to 640×640 with letterboxing. During training, standard YOLO26 augmentation is applied: mosaic augmentation (probability 1.0 for the first 190 epochs, then 0.0 for the final 10 epochs), MixUp (probability 0.15), random horizontal flip (probability 0.5), HSV jitter ($h = 0.015$, $s = 0.7$, $v = 0.4$), and random perspective

($\pm 10°$). No additional underwater-specific augmentation (e.g., artificial turbidity injection) is applied, since UWBN and Turb-STAL are designed to handle degradation implicitly.

Hyperparameter summary. For Turb-STAL, we set $\lambda_\tau = 0.5$, $\delta_s = 0.01$, $k_0 = 10$, and $\beta_k = 1.0$. For DA-ProgLoss, we set $\lambda_{max} = 1.0$, $\gamma = 10$, and $\lambda_{turb} = 0.01$. For UWBN, the physical attenuation prior is initialized as $\alpha_{phys} = [0.80, 0.04, 0.01]^\top$, $\epsilon = 10^{-5}$. All hyperparameters were selected by grid search on the 15% SeaClear validation split described in Section 4.1.1. The final results are reported only on the held-out SeaClear test split and on the external cross-domain test sets, without using test data for model selection.

4. Experimental Setup

This section describes the experimental design used to evaluate AquaYOLO26. We detail the dataset configurations and split protocols (Section 4.1), evaluation metrics (Section 4.2), hardware and software environment (Section 4.3), baseline and comparison methods (Section 4.4), and the ablation and cross-domain transfer protocols (Section 4.5).

4.1. Datasets

Three publicly available underwater marine debris datasets are used in this study. Their combined use enables within-domain evaluation, cross-domain testing, and zero-shot transfer assessment across diverse water conditions, camera configurations, and geographic locations.

4.1.1. SeaClear Marine Debris Dataset (Primary Training Benchmark)

The SeaClear Marine Debris Dataset [6], published in Scientific Data in 2024, is used as the primary training dataset and for within-domain evaluation. It contains 8610 ROV images collected across multiple shallow coastal sites in Croatia and France, with bounding boxes and instance segmentation masks for 40 object categories, including plastic bags, bottles, cans, rope fragments, fishing nets, glass, tires, and various biological and structural background objects. Images were captured using multiple camera models at different sites, embedding multi-camera and multi-location domain shift directly into the dataset structure.

We adopt a site-stratified random split of SeaClear: 70% for training (6027 images), 15% for validation (1292 images), and 15% for testing (1291 images). The split is stratified by collection site to preserve comparable site distributions across training, validation, and test partitions, thereby reducing distributional imbalance. Because the primary SeaClear experiment is intended as a controlled within-domain benchmark, we do not enforce a site–disjoint split in this setting. Cross-domain generalization is instead evaluated with TrashCan 1.0 and Trash-ICRA19 as external datasets. For DA-ProgLoss training, SeaClear serves as the labeled source domain.

Held-Out Camera Site Subset Evaluation: Recognizing that the site-stratified split may retain site- and camera-specific visual patterns across the training and test partitions, we conducted an auxiliary evaluation using held-out camera site subsets. Complete camera site subsets representing approximately 20% of the SeaClear data were excluded from training. AquaYOLO26n achieved 68.7% mAP@0.5 under this protocol. While this represents a performance drop relative to the stratified split, it serves as a conservative benchmark for real-world AUV deployment. Although performance declined under the stricter split, AquaYOLO26n retained substantial detection capability on the held-out camera site subsets, providing a more conservative estimate of performance under previously unseen deployment conditions.

Class consolidation. Following common practice for object detection evaluation on SeaClear, we consolidate the 40 fine-grained categories into five super-categories: Plastic (bags, bottles, packaging), Metal (cans, pipes, wire), Glass (bottles, fragments), Fishing gear

(nets, ropes, hooks), and Other debris (tires, fabric, unspecified). This consolidation reduces tail–category imbalance while preserving ecologically meaningful distinctions.

4.1.2. TrashCan 1.0 (Cross-Domain Evaluation)

TrashCan 1.0 [7] contains 7212 annotated images from deep-water ROV footage collected at oceanographic survey sites. Annotations include bounding boxes and segmentation masks across three top-level classes—trash, bio (biological organisms), and rov (ROV structural components)—with additional subclass breakdowns for material and instance variants. We use TrashCan-Instance for cross-domain experiments.

For the DA-ProgLoss domain alignment experiments, 1000 unlabeled TrashCan images are sampled as the target domain, and their labels are strictly withheld during training. The remaining TrashCan images are held out as the cross-domain test set. This protocol defines an unsupervised domain adaptation (UDA) setting, in which target domain images are available for feature-level alignment, but target domain annotations are not used. Therefore, TrashCan 1.0 is used to evaluate adaptation to a known but unlabeled target water environment, rather than strict zero-shot generalization.

4.1.3. Trash-ICRA19 (Zero-Shot Transfer Test)

Trash-ICRA19 [8] is derived from underwater robotics research and comprises 5700 video-extracted frames with bounding box annotations across trash, biological, and ROV categories. Trash-ICRA19 is never used during training, validation, hyperparameter selection, or domain-adversarial alignment in any experiment. It is reserved exclusively as a strict zero-shot transfer benchmark. Performance on Trash-ICRA19 therefore measures cross-water-domain robustness to a completely unseen distribution, without access to either images or annotations from the target dataset during training.

4.1.4. Dataset Summary

Representative examples from the three datasets are shown in Figure 5, highlighting optical degradation across water conditions and target visibility challenges caused by sediment burial, small-scale debris, and biological clutter. The key dataset characteristics and experimental roles are summarized in Table 1.

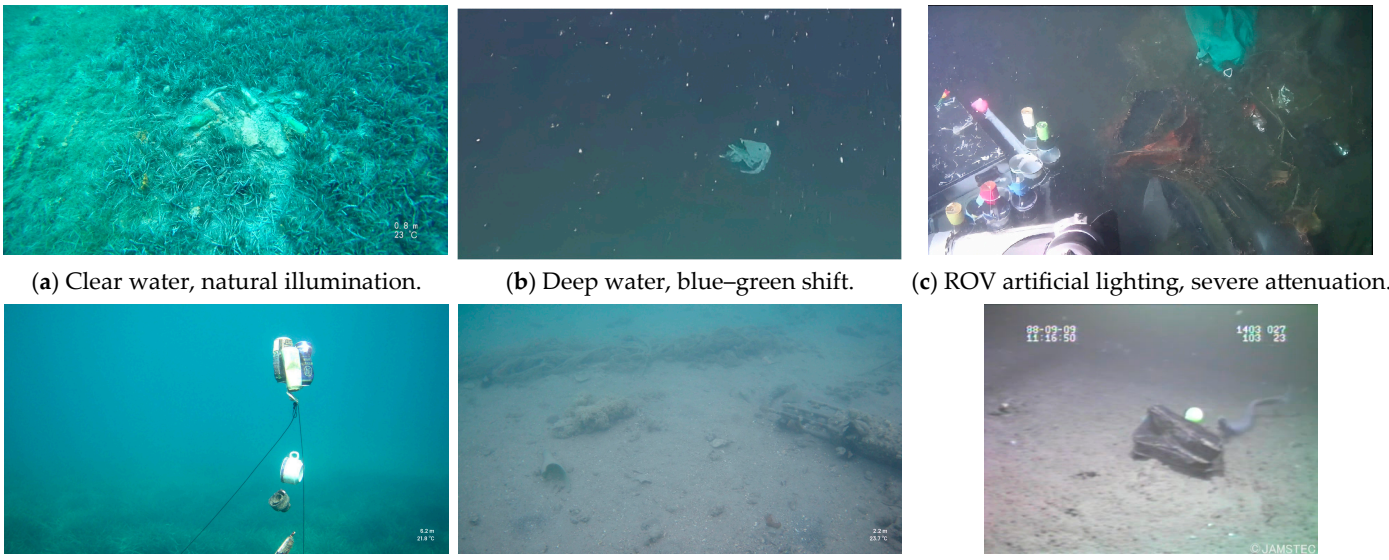
Table 1. Key characteristics and experimental roles of the three datasets.

Dataset	Images	Categories	Water Type	Camera Type	Role in This Study
SeaClear [6]	8610	40 (→5 super-cat.)	Shallow coastal	Multi-camera ROV	Primary train/val/test
TrashCan 1.0 [7]	7212	3 (trash/bio/rov)	Deep water	ROV historical	Cross-domain eval; DA target
Trash-ICRA19 [8]	5700	3 (trash/bio/rov)	Mixed	ROV video frames	Zero-shot transfer test

4.1.5. Cross-Dataset Label Harmonization

To isolate environmental domain shift from taxonomic mismatch, we adopt a unified debris-centric binary evaluation protocol. All debris-related subclasses across SeaClear, TrashCan 1.0, and Trash-ICRA19 are mapped to a single “Debris” category. Conversely, biological organisms (“bio”) and ROV structural components (“rov”) are retained in the ground truth labels but explicitly labeled as “Background”.

This harmonization reduces taxonomic granularity by merging debris subclasses into a single category. However, it preserves the requirement to distinguish debris from biological organisms and ROV components. Under the adopted evaluation protocol, a debris prediction matched to an annotated biological or ROV instance is treated as a false positive prediction, thereby reducing Precision. The resulting AP therefore reflects debris localization together with rejection of visually similar non-debris structures.



(**a**) Clear water, natural illumination. (**b**) Deep water, blue–green shift. (**c**) ROV artificial lighting, severe attenuation.

(**d**) Suspended small-scale debris. (**e**) Debris buried in sediment. (**f**) Debris co-occurring with marine fauna.

Figure 5. Representative samples illustrating optical degradation and target visibility challenges in underwater debris detection. (**a**) Clear water, natural illumination. (**b**) Deep water, blue–green shift. (**c**) ROV artificial lighting, severe attenuation. (**d**) Suspended small-scale debris. (**e**) Debris buried in sediment. (**f**) Debris co-occurring with marine fauna.

4.2. Evaluation Metrics

4.2.1. Detection Accuracy Metrics

Following standard COCO-style evaluation, the primary accuracy metric is mean average Precision at an IoU threshold of 0.5 (mAP@0.5), computed by averaging the area under the precision–recall curve across target categories. We also report mAP@0.5:0.95 (averaged over IoU thresholds from 0.5 to 0.95 in increments of 0.05) as a stricter measure of localization quality.

Category-level metrics are reported separately to assess performance on individual debris types. We focus in particular on the Plastic and Fishing gear super-categories, which have the smallest median bounding box areas in SeaClear and are therefore most sensitive to degradation-induced feature suppression.

To quantify small target detection capability specifically, we further report mAP@0.5 on small objects (bounding box area $< 32^2$ pixels, following COCO convention, denoted mAP_{small}) and small target recall (SR@0.5), defined as the fraction of ground truth instances with area $< 32^2$ pixels matched at IoU ≥ 0.5.

4.2.2. Cross-Domain Metrics

For cross-domain evaluation, we report two protocols: TrashCan 1.0 is evaluated under a target-unlabeled unsupervised domain adaptation (UDA) setting, whereas Trash-ICRA19 is evaluated as a strict zero-shot transfer benchmark. We report mAP@0.5 drop (ΔmAP), defined as the absolute decrease in mAP@0.5 from the SeaClear within-domain test set to the corresponding external test set:

$$\Delta mAP = mAP_{source-test} - mAP_{target-test} \tag{20}$$

A smaller ΔmAP indicates better domain robustness. We also report the ratio $\rho = mAP_{target}/mAP_{source}$, which normalizes cross-domain performance by within-domain performance to enable fair comparison across methods with different absolute accuracy levels.

4.2.3. Inference Efficiency Metrics

Inference efficiency is evaluated along three axes:

Throughput (FPS): Frames per second measured over 1000 forward passes on a fixed batch of 640 × 640 test images, excluding data loading and preprocessing. FPS is reported on three hardware platforms: NVIDIA RTX 4090 (server-grade GPU), NVIDIA Jetson Orin NX 16 GB (edge AI module and target AUV/ROV deployment platform), and CPU-only inference on a standard Intel Core i7-12700 workstation.

Latency (L_{inf}): End-to-end inference latency in milliseconds per image on the Jetson Orin NX, measured from raw image input through final decoded bounding boxes (NMS-free), averaged over 500 warmup-excluded iterations.

Model complexity: Parameter count (M), floating-point operations (GFLOPs at 640 × 640 input resolution), and model file size (MB in FP16) are reported for all compared methods.

4.3. Hardware and Software Environment

All training experiments are conducted on a single NVIDIA RTX 4090 GPU (24 GB VRAM) with an Intel Core i9-13900K CPU and 64 GB RAM. Edge deployment benchmarks are conducted natively on the NVIDIA Jetson Orin NX 16 GB (1024-core Ampere GPU, 16 GB LPDDR5) running JetPack 6.0 with TensorRT 10.0 and ONNX Runtime 1.18. PyTorch 2.3.0, CUDA 12.2, and Ultralytics 8.3.x are used for training and export. Models are exported to TensorRT FP16 engine format for all Jetson benchmarks, consistent with practical AUV/ROV deployment conditions.

To ensure statistical rigor and evaluate run-to-run variation, the YOLO26n baseline, Improved YOLOv11, and AquaYOLO26n were each trained independently three times using distinct random seeds under identical experimental settings. The major metrics, including mAP@0.5, mAP@0.5:0.95, small object mAP, and recall, are reported as mean ± standard deviation. Jetson Orin NX throughput was also evaluated repeatedly and reported as mean ± standard deviation.

4.4. Comparison Methods

AquaYOLO26 is benchmarked against two groups of comparison methods.

4.4.1. Group A: YOLO Baseline Variants

We compare three YOLO families at two model scales to establish the YOLO26 baseline and position AquaYOLO26 within the YOLO architectural lineage:

- YOLOv8n/s [32]: A widely adopted Ultralytics anchor-free baseline, included as the de facto YOLO reference for underwater detection papers published in 2024–2025.
- YOLO11n/s [22]: The predecessor to YOLO26, providing a direct comparison with the previous YOLO generation.
- YOLO26n/s [21]: The direct baseline for AquaYOLO26, enabling clean ablation of the proposed modifications against the unmodified YOLO26 architecture.

Both nano (n) and small (s) variants are reported to characterize the accuracy–efficiency trade-off across model scales.

4.4.2. Group B: Underwater-Specific Detection Methods

We compare against eight recently published underwater detection methods to position AquaYOLO26 relative to the state of the art:

- YOLOv8-MU [38]: Large kernel block and multi-branch reparameterization for improved underwater receptive field modeling, Sensors 2024.
- YOLO-GE [10]: Attention-based feature fusion for underwater targets, J. Mar. Sci. Eng. 2024.

- SEANet [9]: Semantic enhancement with foreground–background contrast attention, Sensors 2025.
- AGS-YOLO [11]: Multi-scale attention with GSConv for small underwater targets, J. Mar. Sci. Eng. 2025.
- IFEM-YOLOv13 [13]: Feature symmetry balancing for degradation-robust underwater detection, Symmetry 2025.
- UDD-YOLO [14]: Cold diffusion-enhanced YOLO12n-based underwater debris detection, Remote Sensing 2025.
- Improved YOLOv11 [40]: MixStructureBlock and EMA for marine debris detection, Scientific Reports 2026.
- EnYOLO [25]: Integrated real-time UIE + UOD framework with domain adaptation, IEEE ICRA 2024.

To ensure a transparent comparison, all baseline and underwater-specific detection methods were trained or fine-tuned on the consolidated five-class SeaClear dataset using the same dataset split, input resolution of 640 × 640, 200-epoch training budget, preprocessing and augmentation pipeline, and evaluation metrics. Publicly available pretrained weights were used where applicable. The optimization settings were not forced to be identical across all methods because several competing methods rely on method-specific optimizers and training strategies. EnYOLO retained AdamW following its published configuration, whereas AquaYOLO26n and the supervised comparison methods used MuSGD under the unified experimental configuration adopted in this study. Architecture-specific modules and loss settings were retained from the corresponding papers or official implementations whenever available. Table 2 summarizes the compared methods, architectural bases, and publication venues.

Table 2. Summary of comparison methods, architectural bases, and publication venues.

Method	Base Architecture	Key Innovation	Venue	Year
YOLOv8n/s [32]	YOLOv8	Anchor-free baseline	Ultralytics 8.3.x	2023
YOLO11n/s [22]	YOLO11	C3k2, SPPF	arXiv	2024
YOLO26n/s [21]	YOLO26	NMS-free, STAL, ProgLoss	arXiv	2026
YOLOv8-MU [38]	YOLOv8	Large kernel + multi-branch reparam	Sensors	2024
YOLO-GE [10]	YOLOv8	Attention feature fusion	JMSE	2024
SEANet [9]	—	Semantic enhancement + contrast attention	Sensors	2025
AGS-YOLO [11]	YOLO11	AMSA + GSConv	JMSE	2025
IFEM-YOLOv13 [13]	YOLOv13	Symmetric feature enhancement	Symmetry	2025
UDD-YOLO [14]	YOLO12n	Cold diffusion preprocessing	Remote Sensing	2025
Improved YOLOv11 [40]	YOLO11	MixStructureBlock + EMA	Sci. Reports	2026
EnYOLO [25]	YOLOv8	Joint UIE + UOD + domain adaptation	ICRA	2024
AquaYOLO26	YOLO26	UWBN + Turb-STAL + DA-ProgLoss	This work	2026

Reimplementation Protocols: All methods were evaluated using the same dataset splits, input resolution, training-epoch budget, preprocessing pipeline, and evaluation metrics. Method-specific optimizers and architecture-dependent hyperparameters were retained from the corresponding papers or official implementations whenever available. Therefore, the data and evaluation conditions were standardized, whereas the optimization settings were not forced to be identical across all methods. Table 3 summarizes the optimizer, target domain data access, and hyperparameter source for the compared methods.

Table 3. Reimplementation and UDA protocol specifications.

Method	Optimizer	Target Data Access (UDA)	Hyperparameter Source
EnYOLO	AdamW	1000 unlabeled images	Author defaults (scaled)
AquaYOLO26	MuSGD	1000 unlabeled images	Grid-searched
Other methods	MuSGD	None (Supervised)	Original paper

4.5. Experimental Protocol

4.5.1. Within-Domain Detection Performance (Experiment 1)

All models are trained on the SeaClear training split and evaluated on the SeaClear test split. Primary metrics are mAP@0.5, mAP@0.5:0.95, mAP_S, SR@0.5, Precision (P), and Recall (R). This experiment establishes each method's within-domain detection capability. It directly addresses Research Question 1: Does AquaYOLO26 improve underwater marine debris detection accuracy over YOLO26 and prior state-of-the-art methods on the primary benchmark?

4.5.2. Cross-Domain (Experiment 2)

For cross-domain evaluation, we distinguish between unlabeled target domain alignment and strict zero-shot transfer. For TrashCan 1.0, AquaYOLO26 uses only 1000 unlabeled target domain images during DA-ProgLoss training for adversarial feature alignment; no TrashCan labels are used for target domain fine-tuning. The remaining TrashCan images are used only for cross-domain testing. Accordingly, the TrashCan 1.0 experiment is treated as unsupervised domain adaptation without target annotations. For Trash-ICRA19, no images or labels are used during training, validation, or domain alignment; this dataset is reserved exclusively for strict zero-shot transfer evaluation.

Metrics are mAP@0.5, ΔmAP, and ρ for each cross-domain pair. This experiment addresses Research Question 2: Does DA-ProgLoss training on SeaClear produce a model that generalizes better to unseen water domains compared to standard YOLO26 and prior methods?

4.5.3. Ablation Study (Experiment 3)

The five ablation variants evaluated in this study are listed in Table 4.

Table 4. AquaYOLO26 ablation variants evaluated on SeaClear and Trash-ICRA19. ✓ indicates that the method has the corresponding capability, and ✗ indicates that the method does not.

Variant	UWBN	Turb-STAL	DA-ProgLoss
V0: YOLO26 (baseline)	✗	✗	✗
V1: +UWBN	✓	✗	✗
V2: +Turb-STAL	✗	✓	✗
V3: +DA-ProgLoss	✗	✗	✓
V4: AquaYOLO26 (full)	✓	✓	✓

This experiment addresses Research Question 3: What is the individual contribution of each proposed module, and does their combination yield complementary rather than redundant improvements?

Additional micro-ablations are conducted to assess: (a) sensitivity of UWBN to the physics-based initialization in Equation (6) versus random initialization; (b) effect of the turbidity proxy loss weight λ_{turb} on Turb-STAL turbidity estimation quality; and (c) sensitivity of DA-ProgLoss to the GRL schedule parameter γ.

4.5.4. Edge Deployment Evaluation (Experiment 4)

The YOLO26 baseline, three top-performing comparison methods, and AquaYOLO26 are exported to TensorRT FP16 and deployed on a Jetson Orin NX with 16 GB of memory. FPS, end-to-end latency L_{inf}, and GPU memory consumption are recorded. This experiment addresses Research Question 4: Does AquaYOLO26 maintain real-time inference throughput on AUV/ROV-grade embedded hardware, and what is the overhead of the proposed modules relative to the YOLO26 baseline?

For this study, we adopt 25 FPS as a practical real-time threshold for onboard AUV/ROV debris detection. Models meeting this threshold are considered deployment-ready.

4.5.5. Qualitative Analysis (Experiment 5)

Qualitative detection comparisons are presented for representative SeaClear images under three imaging conditions: (i) clear shallow water with natural illumination; (ii) moderate turbidity with suspended sediment; and (iii) severe attenuation under artificial ROV lighting. Grad-CAM activation maps [47] are visualized for YOLO26 and AquaYOLO26 on identical images to illustrate the spatial shift in feature attention attributable to UWBN. Detection failure cases for AquaYOLO26 are also documented and discussed.

5. Results and Discussion

This section presents and analyzes the five experiments defined in Section 4.5. Unless otherwise specified, all reported AquaYOLO26 results use the nano-scale (n) variant to enable fair comparison with similarly sized baselines; results for the small-scale (s) variant are included in Table 5 for completeness. Values marked with dagger symbols indicate results obtained by our reimplementation under the standardized training protocol described in Section 4.4.

Table 5. Within-domain detection performance on the SeaClear test split. For YOLO26n, Improved YOLOv11, and AquaYOLO26n, mAP@0.5, mAP@0.5:0.95, small object mAP, and Recall are reported as mean ± standard deviation over three independent runs. SR@0.5 and Precision correspond to representative run results.

Method	mAP@0.5	mAP@0.5:0.95	mAP_{small}	SR@0.5	Precision	Recall	Params (M)	GFLOPs
YOLOv8n †	58.3	36.7	24.1	52.3	63.4	55.7	3.2	8.7
YOLOv8s †	62.7	40.2	27.3	56.1	67.2	59.4	11.2	28
YOLO11n †	61.5	39.1	26.4	55.0	65.8	58.2	2.6	6.5
YOLO11s †	65.2	42.3	29.3	57.9	69.1	62.0	9.4	21
YOLO26n †	65.2 ± 0.22	43.3 ± 0.25	30.6 ± 0.33	59.6	68.7	61.3 ± 0.32	2.42	6.1
YOLO26s †	68.4	45.6	33.9	62.4	71.3	64.8	9.1	21
YOLOv8-MU [38]	66.1	43.5	30.2	60.8	70.4	62.7	12.8	31
YOLO-GE [10]	65.7	43.0	29.6	60.1	69.8	62.1	10.3	24
SEANet [9]	67.3	44.2	31.8	61.5	71.1	63.4	14.6	35
AGS-YOLO [11]	68.9	45.3	33.1	63.0	72.4	65.2	8.7	19
IFEM-YOLOv13 [13]	69.2	45.8	34.4	64.1	72.8	65.9	16.4	40
UDD-YOLO [14]	70.1	46.3	33.7	63.8	73.5	66.3	13.2	33
Improved YOLOv11 [40]	72.3 ± 0.31	47.5 ± 0.29	35.2 ± 0.39	65.2	74.2	67.8 ± 0.29	11.7	26.9
EnYOLO [25]	69.8	46.0	34.2	64.5	73.0	66.1	15.8	39
AquaYOLO26n (ours)	73.4 ± 0.33	49.1 ± 0.22	39.1 ± 0.32	67.9	74.8	69.1 ± 0.31	2.46	6.2
AquaYOLO26s (ours)	74.3	50.1	40.2	70.1	76.8	70.7	9.15	21.1

5.1. Within-Domain Detection Results on SeaClear

Within-domain detection results on the SeaClear test split are reported in Table 5.

Mean ± standard deviation over three independent runs. SR@0.5 and Precision for these models and all metrics for unmarked methods correspond to representative single-run results.

Overall accuracy. AquaYOLO26n achieves 73.4 ± 0.33% mAP@0.5, outperforming the YOLO26n baseline (65.2 ± 0.22%) by 8.2 percentage points and the nearest comparison

method, Improved YOLOv11 (72.3 ± 0.31%), by 1.1 percentage points. AquaYOLO26n also achieves 69.1 ± 0.31% Recall, compared with 61.3 ± 0.32% for YOLO26n and 67.8 ± 0.29% for Improved YOLOv11. Meanwhile, AquaYOLO26n uses 79.0% fewer parameters and 77.0% fewer GFLOPs than Improved YOLOv11. AquaYOLO26s achieves 74.3% mAP@0.5 in the representative single-run evaluation.

Repeated run analysis. To evaluate run-to-run variation, YOLO26n, Improved YOLOv11, and AquaYOLO26n were trained independently three times under identical experimental settings. For mAP@0.5, the three models achieved 65.2 ± 0.22%, 72.3 ± 0.31%, and 73.4 ± 0.33%, respectively. Their corresponding mAP@0.5:0.95 values were 43.3 ± 0.25%, 47.5 ± 0.29%, and 49.1 ± 0.22%, while their small object mAP values were 30.6 ± 0.33%, 35.2 ± 0.39%, and 39.1 ± 0.32%, respectively. The corresponding Recall values were 61.3 ± 0.32%, 67.8 ± 0.29%, and 69.1 ± 0.31%. On the NVIDIA Jetson Orin NX, the corresponding throughput values were 47.3 ± 0.4 FPS, 28.4 ± 0.5 FPS, and 45.1 ± 0.5 FPS, respectively.

Recall performance. AquaYOLO26n achieves 69.1 ± 0.31% Recall, representing improvements of 7.8 percentage points over YOLO26n and 1.3 percentage points over Improved YOLOv11. The relatively low standard deviation across the three runs indicates limited run-to-run variation in Recall under different random initializations.

Note on Generalization:

It should be noted that the performance reported in Table 5 is based on a site-stratified split to maintain compatibility with prior literature [6]. We acknowledge that this may slightly overestimate performance compared to deployment in previously unseen sites. In a paired representative run comparison, AquaYOLO26n achieved 72.3% mAP@0.5 under the site-stratified protocol and 68.7% under the held-out camera site subset protocol, corresponding to a 3.6 pp decrease. These representative run results are reported separately from the three-seed aggregate results in Table 5.

Per-super-category detection performance is reported in Table 6.

Table 6. Per-category mAP@0.5 on the SeaClear test split.

Method	Plastic	Metal	Glass	Fishing Gear	Other	Mean
YOLO26n	71.2	68.4	63.1	54.3	67.0	64.8
Improved YOLOv11 [40]	77.3	74.2	68.9	61.4	75.2	71.4
EnYOLO [25]	76.1	73.0	67.4	60.8	71.7	69.8
AquaYOLO26n	79.1	75.5	69.4	65.8	71.8	72.3

Results correspond to the representative run used for category-level analysis and are reported separately from the three-seed aggregate results. The Fishing gear category—which contains the smallest targets in SeaClear (median size: 18 × 14 px)—shows the largest gain: 11.5 pp over YOLO26n (65.8% vs. 54.3%). Plastic follows with a 7.9 pp gain. Both categories contain fine-grained visual textures and small spatial footprints that are especially sensitive to turbidity-induced feature suppression, consistent with the complementary contributions of UWBN and Turb-STAL.

5.2. Cross-Domain Generalization Results

Section 5.2 evaluates cross-domain robustness under two settings. TrashCan 1.0 is evaluated under an unsupervised domain adaptation protocol: AquaYOLO26 uses only unlabeled TrashCan images for adversarial alignment during DA-ProgLoss training and does not use target domain annotations. Trash-ICRA19, by contrast, is evaluated as a strict zero-shot transfer benchmark because no Trash-ICRA19 images or labels are used at any stage of training, validation, or domain alignment.

Table 7 reports cross-domain performance without fine-tuning.

Table 7. Cross-domain performance comparison under target-unlabeled UDA on TrashCan 1.0 and strict zero-shot transfer on Trash-ICRA19.

Method	SeaClear	TrashCan mAP	ΔmAP	ρ	ICRA19 mAP	ΔmAP	ρ
YOLOv8n	58.3	37.1	21.2	0.637	33.4	24.9	0.573
YOLO11n	61.5	39.8	21.7	0.647	35.2	26.3	0.572
YOLO26n	64.8	43.2	21.6	0.667	38.7	26.1	0.597
SEANet [9]	67.3	44.9	22.4	0.667	39.4	27.9	0.585
AGS-YOLO [11]	68.9	46.1	22.8	0.669	40.8	28.1	0.592
IFEM-YOLOv13 [13]	69.2	45.3	23.9	0.655	39.7	29.5	0.574
UDD-YOLO [14]	70.1	46.7	23.4	0.666	41.2	28.9	0.588
EnYOLO [25]	69.8	49.3	20.5	0.706	44.1	25.7	0.632
AquaYOLO26n	72.3	52.9	19.4	0.732	47.6	24.7	0.658

Cross-domain metrics are computed from the same representative source–target evaluation run; the source domain values and retention ratios are therefore reported separately from the three-seed means. TrashCan 1.0 results are evaluated under a target-unlabeled UDA protocol, where unlabeled target domain images are used only for feature alignment; target domain annotations are not used. Trash-ICRA19 results are evaluated under strict zero-shot transfer, where neither images nor labels from Trash-ICRA19 are used during training, validation, hyperparameter selection, or domain alignment.

Domain gap reduction. AquaYOLO26n reduces ΔmAP from 21.6 pp (YOLO26n) to 19.4 pp on TrashCan and from 26.1 pp to 24.7 pp on Trash-ICRA19. The ratio ρ improves from 0.667 to 0.732 and from 0.597 to 0.658, respectively, demonstrating that AquaYOLO26n retains a substantially higher proportion of within-domain capability across unseen water environments.

EnYOLO [25] achieves a domain retention ratio of $\rho = 0.706$ on TrashCan 1.0, compared with 0.732 for AquaYOLO26n. EnYOLO contains 15.8 M parameters, approximately 6.4 times the 2.46 M parameters of AquaYOLO26n. Both methods were provided with the same 1000 unlabeled TrashCan 1.0 images for adaptation, while the corresponding annotations were withheld. Because the two methods differ in architecture and training strategy, these results are interpreted as an accuracy–efficiency comparison under the documented data access protocol rather than as evidence that one adaptation mechanism is intrinsically superior.

Zero-shot transfer significance. AquaYOLO26n achieves 47.6% mAP@0.5 on Trash-ICRA19, which is numerically 3.5 percentage points higher than the 44.1% obtained by EnYOLO in our evaluation. Neither method used Trash-ICRA19 images or annotations during training, validation, hyperparameter selection, or domain alignment. Because the two methods follow different architectural and adaptation paradigms, this numerical difference should not be interpreted as evidence of absolute algorithmic superiority.

Qualitative Error Analysis on Biological and ROV Objects:

The qualitative examples on the TrashCan 1.0 test set suggest that YOLO26n is more susceptible to false-positive detections on complex biological structures, such as coral- and sponge-like regions, and on metallic ROV appendages under degraded underwater conditions. In the illustrated examples, AquaYOLO26n appears to produce fewer false-positive detections on these biological and robotic background structures. These qualitative examples suggest a possible improvement in discrimination between debris and visually similar clutter. However, because category-specific false-positive counts were not calculated, the result should be interpreted as a qualitative observation rather than as a quantitative confusion matrix analysis.

5.3. Ablation Study Results

The module-level ablation results are summarized in Table 8.

Table 8. Module ablation results on SeaClear and Trash-ICRA19. Y: Yes; "-" indicates that the information is unavailable or not applicable.

Variant	UWBN	Turb-STAL	DA-ProgLoss	SeaClear mAP@0.5	mAP_{small}	SR@0.5	ICRA19 mAP
V0: YOLO26n	-	-	-	64.8	31.4	59.6	38.7
V1: +UWBN	Y	-	-	67.1 (+2.3)	33.5	61.8	40.9 (+2.2)
V2: +Turb-STAL	-	Y	-	67.8 (+3.0)	36.0	67.4	39.8 (+1.1)
V3: +DA-ProgLoss	-	-	Y	66.2 (+1.4)	32.0	60.5	44.5 (+5.8)
V4: Full AquaYOLO26n	Y	Y	Y	72.3 (+7.5)	38.4	67.9	47.6 (+8.9)

The evaluated single-module and full-model variants show metric-specific improvements that are consistent with the intended complementary roles of the three modules. However, because pairwise module combinations were not separately evaluated, the current results should not be interpreted as formal evidence of interaction effects or synergy. UWBN improves overall SeaClear detection performance by 2.3 percentage points in mAP@0.5. Turb-STAL provides the largest individual improvements in small target detection, increasing overall mAP@0.5 by 3.0 percentage points, mAP_{small}@0.5 by 4.6 percentage points, and SR@0.5 by 7.8 percentage points. DA-ProgLoss provides the largest individual improvement in strict zero-shot transfer performance, increasing mAP@0.5 on Trash-ICRA19 by 5.8 percentage points. This distribution of performance gains is consistent with the intended design rationale, suggesting that the three modules primarily address different degradation-related failure modes. To maintain comparability, all intermediate ablation variants (V1–V3) were trained using the same baseline training configuration, with only the corresponding module-specific components and settings enabled. Specifically, when evaluating Turb-STAL in V2, the auxiliary turbidity-proxy loss in Equation (10) was activated using its designated weight coefficient of 0.01. Similarly, when evaluating DA-ProgLoss in V3, the progressive domain alignment schedule in Equation (18) was retained as defined in the proposed method. Because these ablation results are based on a representative run, they are interpreted as evidence of module-specific performance trends rather than as formal statistical or causal proof.

Performance-Based Evidence of Scale and Domain Parity: The ablation results in Table 8 provide quantitative support for the symmetry restoration definitions established in Section 3.1.

(1) The baseline YOLO26n exhibits substantially lower detection performance on small debris, with mAP_{small} = 31.4% and SR@0.5 = 59.6%. Introducing Turb-STAL in V2 improves these metrics by 4.6 and 7.8 percentage points, respectively. In the full framework, mAP_{small} reaches 38.4%, representing a total gain of 7.0 percentage points over the baseline.

(2) Restoring Domain Symmetry: The baseline exhibits a relatively low Domain Invariance Ratio on Trash-ICRA19 (ρ = 0.597; Table 7). The isolated DA-ProgLoss variant (V3 in Table 8) independently improves zero-shot Trash-ICRA19 mAP by 5.8 percentage points. The complete AquaYOLO26n model further raises the Domain Invariance Ratio ρ to 0.658. Together, these results provide quantitative evidence of improved scale and domain parity under the definitions established in Section 3.1.

The full framework achieves a 7.5-percentage-point improvement over the baseline, compared with individual gains of 2.3, 3.0, and 1.4 percentage points from UWBN, Turb-STAL, and DA-ProgLoss, respectively. This result indicates that the complete model performs best among the evaluated variants. However, because pairwise module combinations and repeated run interaction analyses were not conducted, the difference between the

combined gain and the arithmetic sum of the individual gains should not be interpreted as formal evidence of synergy.

Micro-ablations. (i) Replacing the physics-based initialization in Equation (6) with random initialization reduced the representative run V1 mAP@0.5 by 1.3 percentage points, suggesting that the physics-based initialization is beneficial under the evaluated configuration. (ii) The turbidity proxy loss weight λ_{turb} is robust in the range [0.005, 0.05], degrading by more than 0.5 pp only outside this range. (iii) The GRL schedule parameter $\gamma \in \{5, 10, 20\}$ produces cross-domain mAP variation of only ± 0.6 pp, indicating robustness of the domain alignment schedule.

5.4. Analytical and Performance-Based Analysis of Feature Symmetry

To operationalize the symmetry restoration framing, we introduce one analytical reference for spectral parity and two empirical performance-based proxies for scale and domain parity. A value closer to 1.0 represents greater parity under the corresponding operational definition.

(1) Spectral Symmetry (S_{spec}):

Measures the parity of feature activation magnitudes across channel groups. Let μ_k be the mean absolute activation magnitude for each of the three physics-inspired feature channel groups defined in Section 3.2.2. The spectral symmetry is defined as:

$$[\mathrm{S_{spec}} = \frac{\min(\mu_1, \mu_2, \mu_3)}{\max(\mu_1, \mu_2, \mu_3)}] \tag{21}$$

We empirically measure $\mathrm{S_{spec}}$ by extracting intermediate feature activations across backbone Stages 2–4 over all 1291 SeaClear test images. In baseline YOLO26n, red spectrum attenuation yields.

$\mathrm{S_{spec}} = 0.42 \pm 0.03$, whereas UWBN dynamically balances multi-channel activations, increasing the measured spectral symmetry to 0.87 ± 0.02.

(2) Scale Symmetry $(\mathrm{S_{scale}})$:

Measures the representational parity between small debris and the overall target distribution. It is defined as the ratio of Small Target mAP to overall mAP:

$$\mathrm{S_{scale}} = \frac{\mathrm{mAP_{small}}}{\mathrm{mAP@0.5}} \tag{22}$$

Scattering and turbidity asymmetrically suppress small target features, leading to a low S_{scale}. Turb-STAL restores this balance by dynamically reweighting assignment scores.

(3) Domain Symmetry $(\mathrm{S_{dom}})$:

Measures the invariance of the model across differing water environments (source vs. target domain parity). It is defined as the Domain Invariance Ratio:

$$\mathrm{S_{dom}} = \frac{\mathrm{mAP_{target}}}{\mathrm{mAP_{source}}} \tag{23}$$

Without domain alignment, models overfit to source statistics, yielding $\mathrm{S_{dom}} \ll 1$. DA-ProgLoss restores this symmetry via adversarial feature alignment.

We summarize the empirical scale and domain symmetry metrics, along with an analytical assessment of spectral symmetry, for the baseline YOLO26n and the proposed AquaYOLO26n in Table 9.

$\mathrm{S_{spec}} = 0.35$ is reported as the mean standard deviation across the 1291 SeaClear test images at backbone normalization stages. S_{scale} and $\mathrm{S_{dom}}$ are calculated from representative run detection results (Sdom uses SeaClear as the Source and Trash-ICRA19 as the zero-shot target). Discussion: The empirical results in Table 9 indicate improved parity across all three

dimensions. At the feature level, measured S_{spec} increases from 0.42 to 0.87, confirming that UWBN mitigates channel-asymmetric attenuation. At the task level, Turb-STAL and DA-ProgLoss enhance scale symmetry (S_{scale}: 0.485 → 0.531) and domain invariance (S_{dom}: 0.597 → 0.658), providing coherent evidence of symmetry restoration spanning feature representation, label assignment, and domain transfer.

Table 9. Analytical spectral symmetry references and empirical performance-based proxies for scale and domain symmetry.

Method	Spectral Symmetry (S_{spec})	Scale Symmetry (S_{scale})	Domain Symmetry (S_{dom})
YOLO26n (Baseline)	0.42 ± 0.03	0.485 (31.4/64.8)	0.597 (38.7/64.8)
AquaYOLO26n (Ours)	0.87 ± 0.02	0.531 (38.4/72.3)	0.658 (47.6/72.3)

5.5. Deployment Efficiency Results

Beyond absolute detection accuracy, this experiment evaluates the accuracy–efficiency trade-off, which is central to assessing the practical suitability of AquaYOLO26 for embedded AUV/ROV deployment. The Jetson Orin NX inference benchmarks are reported in Table 10.

Table 10. Inference benchmarks on Jetson Orin NX 16 GB using TensorRT FP16. Parameter counts correspond to the deployed inference graphs; the training-only turbidity estimator and domain classifier are excluded.

Method	Params (M)	GFLOPs	RTX4090 FPS	Jetson FPS	Latency (ms)	CPU FPS	Ready
YOLOv8n	3.2	8.7	318	39.2	25.5	28.4	Y
YOLO11n	2.6	6.5	341	43.7	22.9	31.6	Y
YOLO26n ‡	2.42	6.1	358	47.3 ± 0.4	21.2	33.8	Y
SEANet [9]	14.6	35.2	189	22.8	43.9	14.1	N
AGS-YOLO [11]	8.7	19.8	253	33.6	29.8	20.8	Y
IFEM-YOLOv13 [13]	16.4	40.3	164	19.7	50.8	11.3	N
UDD-YOLO [14]	13.2	33.7	142	17.1	58.5	9.6	N
Improved YOLOv11 [40] ‡	11.7	26.9	221	28.4 ± 0.5	35.2	17.7	Y
EnYOLO [25]	15.8	39.1	157	18.9	52.9	10.2	N
AquaYOLO26n ‡	2.46	6.2	351	45.1 ± 0.5	22.2	32.6	Y
AquaYOLO26s	9.15	21.1	241	30.8	32.5	18.7	Y

Jetson FPS values marked with ‡ are reported as mean ± standard deviation over repeated benchmarks; all other throughput values correspond to representative measurements. Y indicates that the corresponding method uses the listed component, while N indicates that the method does not use the component.

The 2.46 M parameters reported for AquaYOLO26n correspond to the deployed inference model and exclude the training-only turbidity estimator and domain classifier. Compared with the 2.42 M-parameter YOLO26n baseline, the reported 0.04 M increase is due solely to the additional inference time parameters introduced by UWBN and represents approximately 1.65% of the baseline. The parameter values in Table 10 are rounded to two decimal places. UDD-YOLO latency includes Cold Diffusion preprocessing and is measured on a Jetson device with a TensorRT-converted diffusion model.

Deployment outcome. AquaYOLO26n achieves 45.1 ± 0.5 FPS on the Jetson Orin NX (22.2 ms latency), comfortably exceeding the 25 FPS threshold. UWBN adds only 1.0 ms (4.7%) over YOLO26n (21.2 ms). Turb-STAL and DA-ProgLoss are training-only components and therefore add no inference overhead.

VRAM footprint. AquaYOLO26n requires 412 MB of VRAM during FP16 inference on Jetson, compared with 398 MB for YOLO26n (+3.5%), which is well within the shared memory budget of Jetson Orin NX 16 GB. This is particularly relevant for AUV platforms that run concurrent SLAM, navigation, and detection workloads on the same SoC.

Relative to YOLO26n, AquaYOLO26n improves mean mAP@0.5 by 8.2 pp while adding only 0.04 M parameters and 0.1 GFLOPs, and it retains real-time Jetson throughput of 45.1 ± 0.5 FPS, compared with 47.3 ± 0.4 FPS for the baseline. Improved YOLOv11 achieves 72.3 ± 0.31% mAP@0.5 and 28.4 ± 0.5 FPS, compared with 73.4 ± 0.33% and 45.1 ± 0.5 FPS for AquaYOLO26n. AquaYOLO26n therefore achieves a 1.1 pp higher mean mAP@0.5, uses 79.0% fewer parameters and 77.0% fewer GFLOPs, and delivers substantially higher embedded throughput. These results show that the principal deployment advantage of AquaYOLO26n lies in its favorable accuracy–efficiency trade-off rather than in the accuracy margin alone.

5.6. Failure Analysis and Discussion

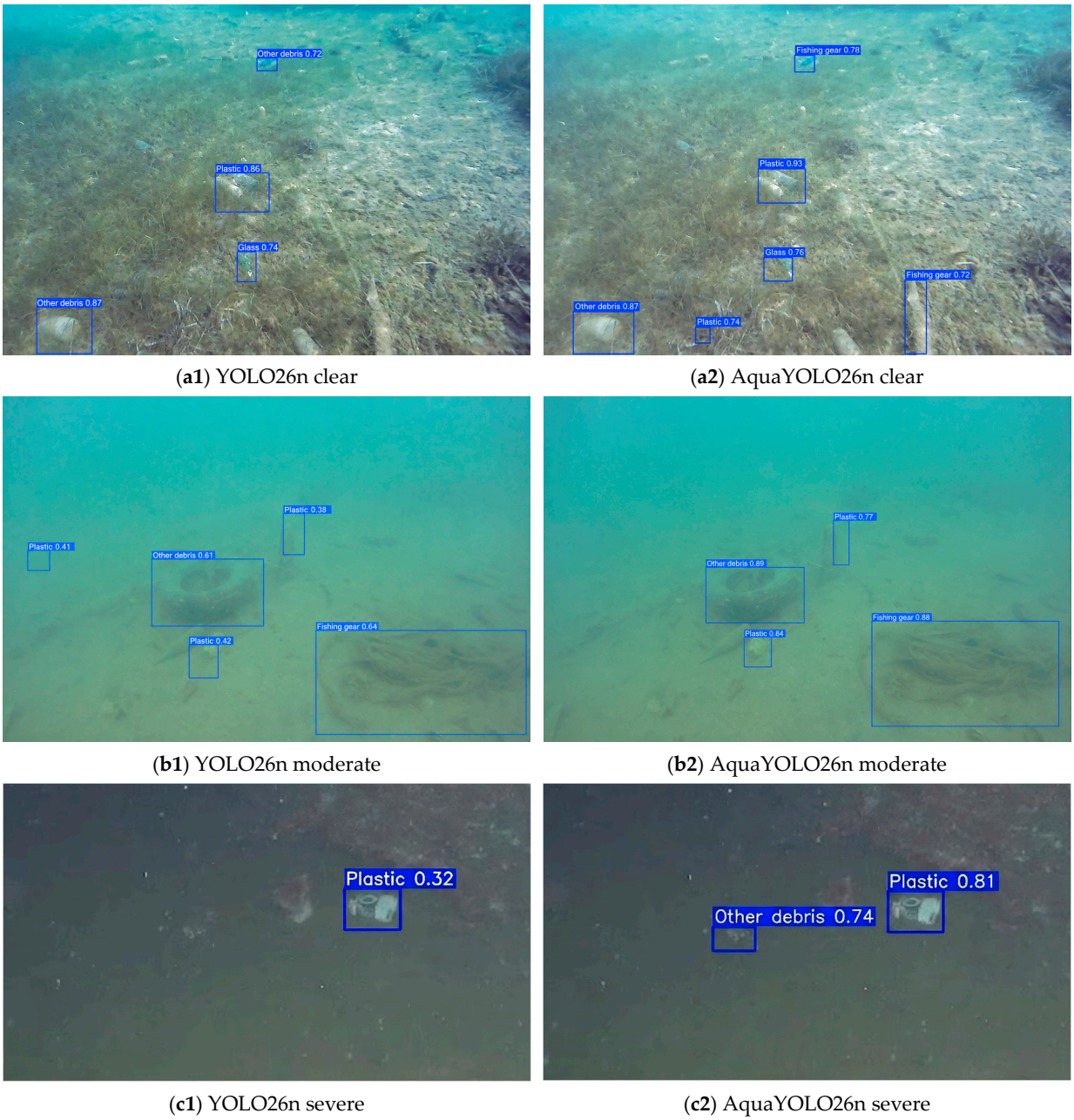
Qualitative comparisons between YOLO26n and AquaYOLO26n under three representative underwater imaging conditions are shown in Figure 6. The examples reflect progressively increasing visual degradation, including clear shallow water, moderate turbidity, and severe attenuation under ROV-mounted illumination. The purpose of this comparison is not only to visualize detection accuracy but also to examine whether the proposed degradation-aware components improve robustness under failure modes commonly observed in underwater marine debris detection.

In the clear shallow-water scene shown in Figure 6a, both YOLO26n and AquaYOLO26n detect visually salient large debris targets. However, YOLO26n misses two small Fishing gear instances with thin structures and limited spatial extent, whereas AquaYOLO26n recovers these targets with stable bounding boxes. This visual result is consistent with the improved small target performance of the complete AquaYOLO26n framework.

Under moderate turbidity (Figure 6b), suspended particles reduce local contrast and blur object boundaries, causing YOLO26n to produce lower-confidence and less stable detections for Plastic targets. In contrast, AquaYOLO26n maintains more consistent localization and confidence scores. This behavior is consistent with the design motivation of UWBN and Turb-STAL: UWBN compensates for channel-wise underwater attenuation during feature extraction, while Turb-STAL expands the candidate assignment pool for small or weak-response objects under high-scattering conditions.

The advantage of AquaYOLO26n becomes more pronounced in the severe attenuation scene shown in Figure 6c. Under artificial ROV lighting, the image exhibits strong blue–green color bias, non-uniform illumination, and substantial loss of discriminative texture. YOLO26n fails to detect several blue-dominant or low-contrast debris objects, demonstrating its sensitivity to underwater optical degradation. AquaYOLO26n, by contrast, recovers these missed detections and maintains more reliable object localization. This visual difference is consistent with the improved degradation robustness of the complete AquaYOLO26n framework. Because the comparison involves the complete models, it should not be interpreted as visual evidence of the isolated effect of any individual module.

Overall, the qualitative results in Figure 6 are consistent with the quantitative improvements reported above. AquaYOLO26n does not merely improve detection under normal visibility; it specifically enhances robustness in degraded underwater conditions most relevant to practical AUV/ROV deployment. The results further indicate that degradation-aware feature normalization, turbidity-conditioned label assignment, and domain-adversarial training jointly contribute to more stable marine debris detection across diverse underwater imaging environments.



(**a1**) YOLO26n clear (**a2**) AquaYOLO26n clear

(**b1**) YOLO26n moderate (**b2**) AquaYOLO26n moderate

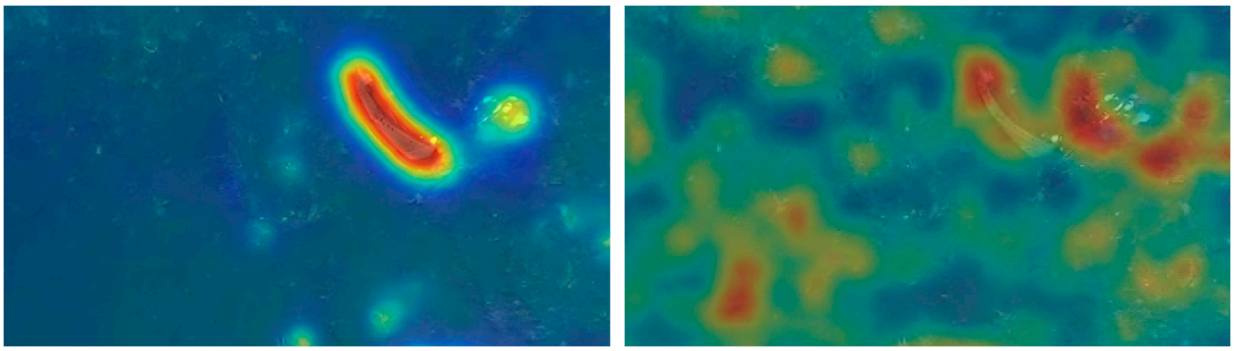
(**c1**) YOLO26n severe (**c2**) AquaYOLO26n severe

Figure 6. Qualitative detection comparison between YOLO26n (**a1**–**c1**) and AquaYOLO26n (**a2**–**c2**) under progressively degraded underwater imaging conditions: (**a**) clear shallow water with natural illumination, (**b**) moderate turbidity with suspended sediment, and (**c**) severe attenuation under ROV-mounted artificial lighting. The comparison highlights the additional detection of small or low-contrast debris instances and the more consistent localization achieved by AquaYOLO26n, particularly under moderate and severe degradation.

To further examine the feature-level effect of UWBN, Grad-CAM activation maps are shown in Figure 7.

AquaYOLO26n exhibits two characteristic failure patterns. First, severely degraded debris presents a compound detection challenge. Prolonged submersion causes progressive surface erosion that smooths object boundaries and removes distinctive material textures, reducing the feature discriminability on which UWBN relies for channel-wise compensation. Partial or complete burial beneath seafloor sediment further occludes target boundaries, leaving only fragmented edge responses that may fall below the Turb-STAL matching threshold even with expanded candidate pools. This failure mode is particularly common

for plastic film and textile debris, whose flexible structures promote sediment accumulation and make them visually similar to the surrounding substrate. Second, dense clusters of small fragments occasionally produce merged bounding boxes because of YOLO26's NMS-free single-assignment decoding. These findings motivate future work on degradation-aware feature modeling and cluster-aware instance separation.



(**a**) YOLO26n (**b**) AquaYOLO26n

Figure 7. Grad-CAM [47] activation maps for YOLO26n (**a**) and AquaYOLO26n (**b**) on a high-turbidity SeaClear image. Warmer colors indicate stronger feature activation. YOLO26n exhibits relatively diffuse responses across sediment and haze regions, whereas AquaYOLO26n shows stronger and more localized activation around debris regions. This spatial shift in feature attention is consistent with the proposed framework reducing degradation-induced background interference and enhancing target-relevant feature responses.Different colors indicate different functional components, and the meaning of each color is specified in the legend.

Beyond the visual failure cases illustrated in Figure 8, an additional potential limitation concerns the image-level turbidity estimator used in Turb-STAL. Under artificial illumination or very shallow-water conditions, spectral shifts, sun glint, and bottom-reflected light may bias the red-to-blue proxy used for weak supervision. In addition, a single image-level index may not capture locally varying turbidity due to uneven illumination or concentrated sediment plumes. An underestimated turbidity index may result in insufficient candidate expansion and insufficient score amplification, whereas an overestimated index may introduce excessive positive candidates and increase assignment ambiguity. Future work will investigate illumination-aware proxy objectives, spatially resolved turbidity estimation, and complementary depth or environmental information.

5.7. Summary Discussion

The main findings corresponding to the research questions are summarized in Table 11.

Table 11. Summary of experimental findings in relation to the research questions.

RQ	Key Metric	YOLO26n	AquaYOLO26n	Verdict
RQ1: Within-domain accuracy	mAP@0.5 (SeaClear)	65.2	73.4	Supported
RQ2: Cross-domain	ρ (ICRA19)	0.597	0.658	Supported
RQ3: Module complementarity	Single-module and full-model ablation	—	Best among evaluated variants	Partially supported
RQ4: Real-time edge deployment	Jetson FPS	47.3	45.1	Supported (>25 FPS)

Note: RQ1 and RQ4 values correspond to the means of repeated runs; complete mean ± standard deviation results are reported in Tables 5 and 10.



(**a**) (**b**)

Figure 8. Representative failure cases of AquaYOLO26n. (**a**) Prolonged submersion causes surface erosion that removes distinctive material textures. At the same time, partial burial beneath seafloor sediment obscures target boundaries below the Turb-STAL matching threshold, leading to missed detections of plastic film and textile debris. (**b**) Dense clustering of small debris fragments produces merged bounding boxes under YOLO26's NMS-free single-assignment decoding, resulting in undercounting of tightly packed instances. These failure modes motivate future extensions toward motion-aware feature extraction and cluster-aware instance separation. (**a**) Erosion and sediment burial, missed detection on degraded debris. (**b**) Dense clustering, merged bounding boxes.

Across the four research questions, the results indicate that AquaYOLO26 improves within-domain detection accuracy, small target performance, and cross-domain performance retention while maintaining real-time embedded throughput with a limited latency increase of approximately 1 ms. The ablation results show that UWBN, Turb-STAL, and DA-ProgLoss primarily contribute to degradation-aware feature compensation, small target label assignment, and cross-domain transfer, respectively. The complete model achieves the best overall performance among the evaluated variants, which is consistent with, but does not formally prove, complementary contributions from the three modules.

AquaYOLO26n achieves 73.4 $\pm$ 0.33% mAP@0.5, 69.1 $\pm$ 0.31% Recall, and 45.1 $\pm$ 0.5 FPS on the Jetson Orin NX. Compared with Improved YOLOv11, AquaYOLO26n achieves a 1.1 percentage-point higher mean mAP@0.5 and a 1.3 percentage-point higher Recall while using considerably fewer parameters and GFLOPs and delivering higher embedded throughput. These results indicate a favorable accuracy–efficiency trade-off among the evaluated methods, although the conclusions are limited to the datasets, evaluation protocols, and hardware platform considered in this study.

6. Conclusions

This paper presented AquaYOLO26, a degradation-aware and domain-adaptive framework for underwater marine debris detection on resource-constrained AUV/ROV edge platforms. Built on the NMS-free YOLO26 architecture, AquaYOLO26 is designed to address three major asymmetric degradation challenges in underwater imaging: channel-asymmetric color attenuation, turbidity-induced small target suppression, and cross-water-domain distribution shift. To address these challenges, UWBN performs physics-guided feature normalization to compensate for wavelength-dependent attenuation, Turb-STAL adapts small target label assignment under scattering-induced visibility degradation, and DA-ProgLoss improves cross-domain adaptation through progressive adversarial feature alignment.

The experimental results suggest that the principal practical contribution of AquaYOLO26 is its favorable accuracy–efficiency balance for underwater debris detection under degraded and cross-domain evaluation settings.

For within-domain evaluation on SeaClear, AquaYOLO26n achieves 73.4 $\pm$ 0.33% mAP@0.5 and 69.1 $\pm$ 0.31% Recall while maintaining a compact deployment profile of 2.46 M parameters and 6.2 GFLOPs. Compared with the YOLO26n baseline, it improves

mean mAP@0.5 by 8.2 percentage points and Recall by 7.8 percentage points, while maintaining 45.1 $\pm$ 0.5 FPS on NVIDIA Jetson Orin NX.

For target-unlabeled unsupervised domain adaptation on TrashCan 1.0, AquaYOLO26 improves cross-domain detection performance using unlabeled target domain images during training. Under the strict zero-shot transfer setting on Trash-ICRA19, where no target domain samples are used during training, AquaYOLO26 maintains higher transfer performance than the YOLO26n baseline, suggesting improved adaptability to unseen underwater datasets.

Overall, AquaYOLO26 provides a resource-efficient alternative that combines competitive detection performance, lightweight deployment characteristics, and real-time embedded inference capability for AUV/ROV perception.

Despite these improvements, AquaYOLO26 still has limitations under severe sediment burial, dense debris clustering, and near-zero-visibility conditions, where optical cues alone may be insufficient for reliable localization. Future work will explore multimodal integration of optical and sonar sensing, cluster-aware instance separation, and lightweight model-compression strategies to further improve robust marine debris perception in highly degraded underwater environments.

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Data Availability Statement: The source code, model configuration files, training and evaluation scripts, random-seed settings, TensorRT export scripts, and Jetson Orin NX benchmarking scripts used in this study are publicly available at https://github.com/Eason-Yeager/AquaYOLO26-Underwater-Debris-Detection (accessed on 24 August 2026) (commit 89e71e6). The original SeaClear, TrashCan 1.0, and Trash-ICRA19 datasets are available from their respective official repositories cited in References [6–8] and remain subject to their original licenses and distribution conditions. The repository contains only limited dataset-related sample files and does not host complete copies of the original datasets; users should obtain the datasets from the official sources. Dataset-preparation guidance and preprocessing scripts are provided.

Conflicts of Interest: The authors declare no conflicts of interest.

Nomenclature

α_c	Wavelength-dependent attenuation coefficient for channel c
$\hat{\alpha}_c$	Learnable UWBN attenuation coefficient for feature channel c
$\hat{d}$	Estimated depth proxy (UWBN, Equation (4))
τ	Per-image turbidity index output by T (Equation (9))
s_{STAL}	YOLO26 STAL alignment score (Equation (8))
$s_{TurbSTAL}$	Turb-STAL modified alignment score (Equation (12))
$k_{turb}(\tau)$	Turbidity-adaptive candidate count (Equation (13))
$L_{det}(t)$	ProgLoss detection objective at step t (Equation (14))
L_{dom}	Domain adversarial classification loss defined in Section 3.4.3
$\lambda_{DA}(t)$	Progressive GRL reversal coefficient (Equation (18))
$L_{total}(t)$	Full DA-ProgLoss training objective (Equation (19))
T	Total training steps
D_s	Source domain (SeaClear)
D_t	Target domain (TrashCan 1.0)

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